

Research paper

Simulation study of dynamic building insulation with transpired solar collectors

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ABSTRACT

This study conducts a comprehensive numerical analysis using Ansys Fluent to investigate the thermal behaviour of transpired solar collectors (TSCs) focusing on temperature distribution, airflow dynamics, and heat recapture capabilities. The research evaluates TSC performance across various environmental conditions. A significant observation from the study is the TSCs' ability to recapture heat losses from the enclosures during nighttime, particularly under higher airflow conditions, where TSCs effectively redirect heat back into building spaces, contributing to energy conservation. Subsequently, the study reveals an inverse relationship between solar radiation levels and TSC efficiency in heat loss recapture. As solar radiation increases from 200 W/m² to 1000 W/m², the recaptured wall heat loss decreases by a percentage, ranging from 125 % to 241 % compared to the maximum observed during nocturnal periods. Additionally, the temperature analysis of the TSC backplate demonstrates a complex thermal response under different operational conditions. Solar radiation, airflow rates, and environmental properties influence this interplay. The study also uncovers a consistent linear increase in airflow velocity from the inlet to the outlet within the TSC air cavity, suggesting efficient airflow management. This insight is crucial for optimizing TSC design for improved heat exchange efficiency. In conclusion, the findings of this research offer valuable insights into the operational dynamics of TSCs, highlighting the importance of environmental factors in their design and functionality.

1. Introduction

Integrating cutting-edge technologies and sustainable practices in modern building engineering is crucial for promoting energy-efficient construction (Saidi and Hammami, 2015; Dabbas et al., 2021). Solar heat utilisation is a well-developed and cost-effective method, particularly concerning buildings (Filimonova et al., 2021). Among various solar technologies, transpired solar collectors (TSC), also known as solar walls, have gained attention due to their compatibility with building facades (Pop et al., 2022). These collectors can be integrated into a building's exterior to heat incoming air, meeting its energy needs. The concept of TSC is relatively new, with its development dating back to the mid-1980s by John Hollick and Rolf Peter and its first commercial application in the early 1990s at the Ontario Ford Motor Company assembly factory (Hollick, 1994).

To create a TSC, a perforated metallic sheet that absorbs solar energy is attached to a building's outer surface, leaving an air gap or plenum between the sheet and the building. A fan draws external air into this plenum through evenly spaced holes in the absorber's surface. As the air passes through the sheet, it absorbs heat through convection (Eryener and Kescu, 2018; Eryener et al., 2017). The study of TSC involved experimental, analytical, and numerical methods, including computational fluid dynamics (CFD) simulations, which started in the late 1990s and initially focused on two-dimensional models (Kutscher, 1992, 1997; Gawlik, 1995).

Dynamic insulation involves adjusting a building's thermal resistance (R-value) to regulate heat transfer into and out of the structure. This approach can be integrated into the building's structure as a heat exchanger, preheating outdoor air in winter. This controlled adjustment of the envelope's R-value offers significant energy efficiency benefits. It regulates heat transfer through the building's walls, roof, and floors,

Abbreviations: TSC, Transpired solar collector; CFD, Computational fluid dynamics; UTSCs, Unglazed transpired solar collectors; ASHRAE, American Society of Heating, Refrigerating and Air-Conditioning Engineers; IEA, International Energy Agency.

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Nomenclature

Latin characters

$\dot{Q}_{\text{sol}}$	Solar irradiance incident on the collector plate [W]
$\dot{Q}_{\text{conv, coll-cavity}}$	Convective heat transfer from the TSC collector to the air cavity [W]
$\dot{Q}_{\text{conv, coll-amb}}$	Convective heat transfer from the TSC collector to the ambient [W]
$\dot{Q}_{\text{rad, coll-surr}}$	Radiative heat transfer between the collector and the surroundings [W]
$\dot{Q}_{\text{rad, coll-back}}$	Radiative heat transfer between the TSC collector and the backplate [W]
$\dot{Q}_{\text{conv, back-cavity}}$	Convective heat transfer from the TSC backplate to the air cavity [W]
$\dot{Q}_{\text{conv, in-back}}$	Convective heat transfer between the building interior and the backplate [W]
$\dot{Q}_{\text{rad, in-back}}$	Radiative heat transfer between the building interior and the backplate [W]
G_{coll}	Total solar irradiance incident on the collector [W]
T_{coll}	Temperature of the collector [K]
T_{cavity}	TSC air cavity (plenum) temperature [K]
T_{amb}	Ambient air temperature [K]
$F_{\text{coll-sky}}$	View factor from the collector to the sky [1]
T_{sky}	Effective sky temperature [K]
$F_{\text{coll-gnd}}$	View factor from the collector to the ground [1]
T_{gnd}	Ground temperature [K]
T_{back}	TSC backplate temperature [K]
h_{back}	Convective heat transfer coefficient of the backplate [W/(m ² ·K)]
T_{in}	Indoor (interior) average temperature [K]

T_{out}	Temperature of the air leaving the cavity [K]
h_{coll}	Convective heat transfer coefficient of the TSC collector [W/(m ² ·K)]
$\dot{V}$	Airflow rate produced by the fan [m ³ /m ² ·h]
c_p	Specific heat capacity [kJ/(kg K)]
R	Building envelope thermal resistance [(m ² ·K)/W]
A_{coll}	The effective surface area of the TSC collector [m ²]
A_{back}	The effective surface area of the TSC backplate [m ²]
f	The porosity of the TSC absorber plate [1]
D	Perforation diameter [m]
P	Pitch (centre-to-centre distance) of the perforations [m]
k	Perforated plate loss coefficient [1]
C_1	Hole centre-to-centre distance in the horizontal direction [1]
C_2	Hole center-to-center distance in the vertical direction [1]
ΔP	Total pressure drop across the collector [Pa]
v	Plenum (cavity) air velocity [m/s]
L	Length [m]
W	Width [m]
x	Thickness [m]

Greek letters

η	Collector efficiency [1]
ε	Collector effectiveness [1]
α	Solar absorptivity of the solar collector [1]
ρ	Density [kg/m ³]
λ	Thermal conductivity [W/m·K]
ϵ_{coll}	Collector emissivity [1]
σ	Stefan-Boltzmann constant [W/m ² ·K ⁴]
τ	Time [s]

reducing energy consumption and costs. Accordingly, increasing the R-value prevents heat loss, keeping indoor spaces warm with reduced heating system usage. This adaptability in R-value management promotes year-round energy efficiency, cost savings, and a comfortable indoor environment while contributing to reduced greenhouse gas emissions and eco-friendly building design (Fawaier and Bokor, 2022).

The effectiveness of dynamic insulation is influenced by several factors, including the building structure, climate conditions, envelope materials, and physical phenomena within building facades, such as heat transfer and airflow velocity (Koenders, 2018). In-depth research is essential to explore these aspects and harness the potential of dynamic insulation in reducing building energy consumption (Appiah et al., 2023).

Accordingly, dynamic insulation in building construction is a promising field that demands further research. Addressing material choices, permeability, and understanding heat transfer and airflow within wall cavities are critical aspects (Imbabi, 2006; Elsarrag et al., 2012; Taylor et al., 1998). With thorough research and development, dynamic insulation technology promises significant energy savings.

Although perforated plates are not a relatively new technology, there is a lack of research describing their heat loss (Fawaier and Bokor, 2020, 2021). Numerous research studies have highlighted the additional advantages of reducing heat loss through transmission by up to 50 % through heat recirculation (Balázs et al., 2019; NRCan, 2005). During the winter of 1993/94, a comprehensive case study was conducted at the General Motors battery plant in Oshawa, Canada (Federal Technology Alert, 1998). This study employed a transpired solar collector with an area of 420 m² for ventilation. The analysis of measured data revealed that approximately 38 % of the annual energy savings could be attributed to the recapture of heat from transmission loss. It is noteworthy that

Li et al. (2014). specifically highlighted the oversight in previous studies concerning neglecting recaptured heat flux within the context of the transpired solar collectors. As a result, a research gap exists in this particular aspect of TSC literature.

Moreover, researchers had primarily focused on flat perforated types. Gawlik and Kutscher (2002) and Gawlik et al. (2005) conducted experimental and numerical work, considering system aspects such as geometry, absorber porosity, wind, and suction velocity. Subsequently, studies were undertaken at the University of Waterloo by Cao et al. (1993), Golneshan (1996), Gunnewiek et al. (1996), Arulanandam et al. (1999), and Van Decker et al. (2001). These studies also encompassed experimental and numerical approaches and focused on designing and developing the transpired air collector.

The earliest work dedicated to unglazed transpired solar collectors was conducted by Kutscher et al. (1993a, 1993b). They considered a flat plate absorber with homogeneous suction. Initially, Kutscher studied convective heat loss from the TSC under no-wind (natural convection) conditions. When homogeneous suction and laminar flow over the absorber are assumed, analytical results demonstrate that the flow over the absorber plate becomes asymptotic after a certain starting length. The asymptotic region has no net convective heat flux because the boundary layer is drawn through the perforations, preventing further boundary layer growth. Thus, convective heat losses only occur in the starting length.

Ultimately, it could be noted that previous studies have independently explored the effectiveness of dynamic building insulation and transpired solar collectors. A substantial body of literature showcases their advantages in mitigating energy demands and optimizing building thermal conditions. However, there is a clear gap in the current research because there has been limited investigation into how dynamic building

insulation and transpired solar collectors can be effectively combined (Fawaier et al., 2024). While both technologies have proven their worth individually, their coherent application remains largely unexplored.

This study aims to develop a high-resolution numerical model to accurately describe the heat transfer process when using TSCs as dynamic insulation. It must be noted that the whole TSC heat production consists of two parts: the first one is heating the exterior air using the solar radiation incident on the perforated plate. The second part represents that when TSC is in operation mode, it can recapture the transmission heat losses inside the TSC air cavity. The study's objectives include analysing the heat transfer characteristics of TSC as dynamic insulation and evaluating its effectiveness in reducing energy consumption in buildings.

Accordingly, Fig. 1 illustrates the algorithmic process used in the study to evaluate the performance of TSC using computational fluid dynamics. The process advances through several computational stages, starting with the initial inputs, including ambient conditions, operational parameters, and TSC geometry/materials. These stages encompass setting up the CFD model, performing simulations, and post-processing the results to extract critical data on temperature distribution, heat flux profiles, and other relevant performance metrics. The final outputs provide insights into the thermal performance of the TSC, offering guidelines for design optimization and operational efficiency.

2. Numerical model

2.1. Collector geometry

A three-dimensional CFD model has been developed using Ansys Fluent software (version 2020 R1) (Fluent, 2020). The resulting numerical model simulates the implementation of TSC in real-life scenarios and how it would behave as a dynamic insulator. The study aims to provide a high-resolution mesh to accurately describe the heat transfer process using TSC as active insulation. Fig. 2 indicates the schematic representation of ANSYS CFD geometry, illustrating the front view (a) and back view (b). TSC collectors are designed vertically and comprise two primary components: a perforated absorber and a back plate. An air gap, the plenum, is maintained between these components. Enclosing the gaps along the absorber's edges forms a closed-box structure, with a blower situated at the uppermost part of the box to facilitate the required suction during collector operation.

The front element of the TSC design is a semi-transparent perforated plate. This plate permits a significant portion of incident solar radiation

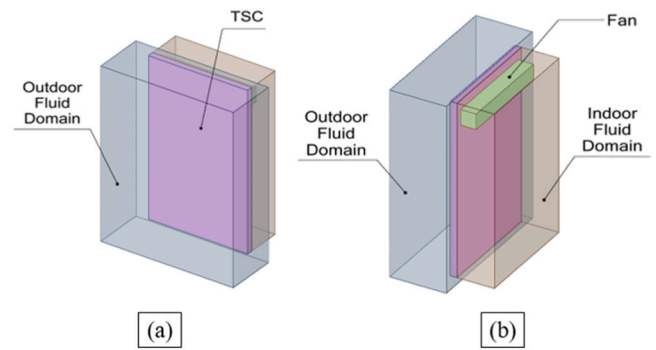


Fig. 2. Schematic representation of ANSYS CFD geometry, illustrating the tri-metric front view (a) and back view (b).

(irradiation) to transmit to the internal absorber wall while possessing high transmissivity and low absorptivity to solar radiation on its outer surface. This feature limits the cover plate's temperature increase, effectively mitigating heat losses to the surrounding environment via radiation and convection. The conjunction of the transparent cover plate with the high-absorptivity absorber wall enables the capture of radiation. The cover plate's low transmissivity in the near-infrared spectrum ensures that radiation emitted by the inner wall is predominantly absorbed and reflected into the collector.

To facilitate airflow through the perforations in the plate, fans positioned at each end of the absorber wall create a pressure differential within the plenum. As air traverses the perforated plate, it experiences heating through convection. Subsequently, the air continues to gain thermal energy as it interacts with the absorber's inner wall until it reaches the intake ducts.

2.2. Assumptions

This section outlines the foundational assumptions for the transpired solar collectors numerical model. These assumptions simplify complex realities, providing a basis for the mathematical model's insights into TSC performance. The assumptions are presented transparently to clarify the foundation of the analyses and conclusions.

- Steady-state conditions are employed to simplify the mathematical representation of the system, enabling a more manageable numerical

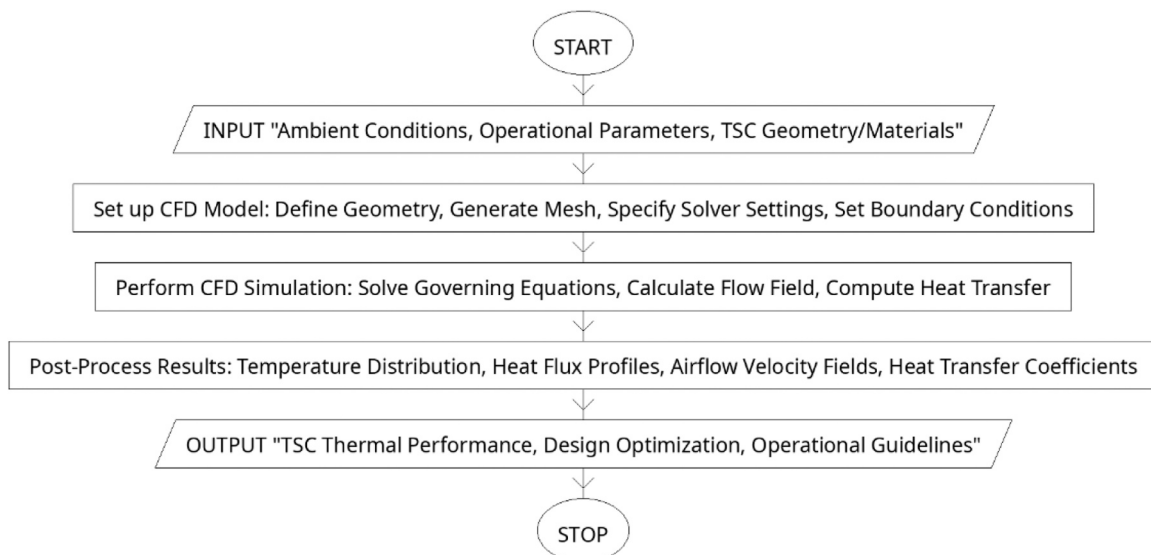


Fig. 1. Flowchart illustrating the step-by-step process of the CFD simulation used to analyse the thermal dynamics of TSCs.

solution. This assumption is justified by the relatively slow variations in solar collector performance under typical operating conditions (Kutscher, 1992; Dymond, 2023).

- According to Kutscher et al. (2003), to mitigate the impact of wind, the following specific conditions are taken into account:

1. In order to accomplish optimal efficiency, it is advisable to maintain a minimum suction face velocity of 0.02 m/s, with a preferable range falling between 0.04 and 0.05 m/s.
2. The selection of wall porosity is critical to ensure an adequate pressure drop across the wall, and it is recommended to maintain a minimum of 25 Pa.

- Optimizing the perforation pattern on the TSC to ensure uniform airflow is critical for consistent thermal performance (Gawlik, 1995; Kutscher et al., 1993b).
- Flow reversal through the absorber is assumed to be negligible. This assumption is based on recommendations to minimize flow reversal effects by maintaining a minimum pressure drop across the absorber plate (Kutscher, 1994; Barker and Kiatreungwattana, 1997).
- Fluid properties are considered constant within the specified temperature range. This assumption is based on the slight temperature differences encountered in the system (Collins and Abulkhair, 2014).
- The influence of gravity on the system is considered negligible. This assumption is supported by research demonstrating that losses by natural convection under no-wind conditions were on the order of 0.1 % of the total heat transfer, indicating minimal gravitational effects (Kutscher et al., 1993a).

Additionally, other potential factors were disregarded in this derivation to keep the model more tractable as it focuses on developing the model of the TSC backplate, such as:

- Factors like system leaks within the cavity air might also influence the cavity's temperature. Still, these effects might be minor compared to the dominant ones mentioned above.
- There is potential for condensation or moisture effects, especially if the TSC cools the interior air below the dew point.
- The potential temperature stratification within the TSC or room would make the interior temperature (T_{in}) non-uniform.

2.3. Energy Balance Equations

In the current analysis, the energy balance for the transpired solar collector is defined as incorporating three elements: the backplate, the air cavity, and the perforated absorber plate. This balance considers the solar input and the heat and mass flow rates across the collector, as shown in Fig. 3.

2.3.1. Absorber Plate Energy Balance

The absorber plate is the frontline for solar energy capture in a TSC. Its efficiency depends on how the absorbed solar energy interacts with the environment and the air cavity dynamics. The energy balance at the absorber plate is critical for understanding how effectively the TSC converts incident solar energy into usable heat. This balance is articulated as follows:

$$\dot{Q}_{sol} = \dot{Q}_{conv,coll-cavity} + \dot{Q}_{conv,coll-amb} + \dot{Q}_{rad,coll-surr} + \dot{Q}_{rad,coll-back} \quad (1)$$

After expanding the terms, the equation would be:

$$\alpha \cdot G_{coll} \cdot A_{coll} = \left[\begin{aligned} &h_{coll} \cdot A_{coll} \cdot (T_{coll} - T_{cavity}) + h_{coll} \cdot A_{coll} \cdot (T_{coll} - T_{amb}) \\ &+ \sigma \cdot A_{coll} \cdot \epsilon_{coll} \cdot (T_{coll}^4 - F_{coll-sky} \cdot T_{sky}^4 - F_{coll-gnd} \cdot T_{gnd}^4) \\ &+ \sigma \cdot A_{coll} \cdot \left(\frac{T_{coll}^4 - T_{back}^4}{\frac{1}{\epsilon_{coll}} + \frac{1}{\epsilon_{back}} - 1} \right) \end{aligned} \right] \quad (2)$$

The value of (α) represents the solar absorptance, a dimensionless value between 0 and 1, indicating the fraction of solar radiation absorbed by the surface. A value of 1 means complete absorption, while 0 means complete reflection. The literature recommended a value of $\alpha \geq 0.9$ for optimal thermal performance (Van Decker et al., 2001).

2.3.2. Backplate energy balance

The backplate functions as a thermal interface, recycling heat from the interior back into the system. Its role extends beyond mere separation; it is an active component in the TSC's thermal cycle. The energy balance for the backplate is described as:

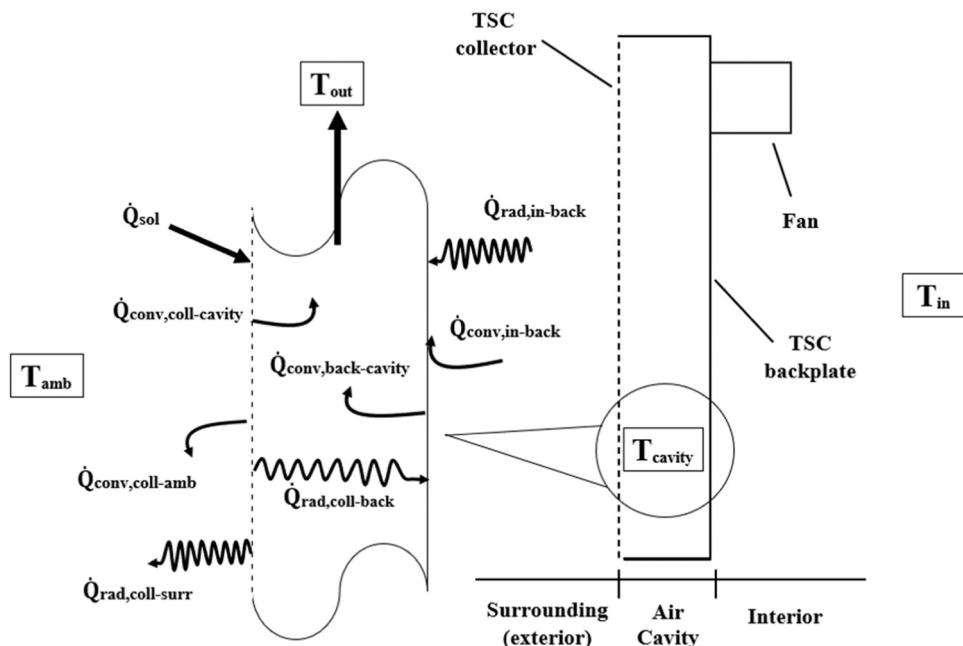


Fig. 3. Heat transfer mechanisms within the transpired solar collector air cavity focusing on the collector backplate.

$$\dot{Q}_{conv,in-back} + \dot{Q}_{rad,in-back} + \dot{Q}_{rad,coll-back} = \dot{Q}_{conv,back-cavity} \quad (3)$$

Expanding the terms:

$$\left[\begin{aligned} &h_{back} \cdot A_{back} \cdot (T_{in} - T_{back}) \\ &+ \sigma \cdot A_{back} \cdot \epsilon_{back} \cdot (T_{in}^4 - T_{back}^4) \\ &+ \sigma \cdot A_{back} \cdot \left(\frac{T_{abs}^4 - T_{back}^4}{\frac{1}{\epsilon_{coll}} + \frac{1}{\epsilon_{back}} - 1} \right) \end{aligned} \right] = h_{back} \cdot A_{back} \cdot (T_{back} - T_{cavity}) \quad (4)$$

The energy balance for the backplate involves convective and radiative heat transfer mechanisms influenced by temperature differentials and surface characteristics. Convective heat transfer from the interior space to the air cavity and radiative heat exchange with the interior space and the absorber plate are crucial factors in the backplate's energy balance.

2.3.3. Air cavity energy balance

The air cavity is where the bulk of heat transfer to the air stream occurs. The dynamics within this cavity are complex, influenced by both the absorber plate and the backplate. The energy balance here can be described by:

$$\dot{Q}_{conv,coll-cavity} + \dot{Q}_{conv,back-cavity} = \dot{V} \cdot \rho_{air} \cdot c_{p,air} \cdot (T_{out} - T_{cavity}) \quad (5)$$

By expanding and integrating the individual terms, the equation would be as follows:

$$\begin{aligned} &h_{coll} \cdot A_{coll} \cdot (T_{coll} - T_{cavity}) + h_{back} \cdot A_{back} \cdot (T_{back} - T_{cavity}) \\ &= \dot{V} \cdot \rho_{air} \cdot c_{p,air} \cdot (T_{out} - T_{cavity}) \end{aligned} \quad (6)$$

As the derivation progresses, it is crucial to begin rearranging the energy balance equation for the backplate, considering all previously derived components and equations. The inability to eliminate T_{back} from the right side of the equation arises from the equation's inherent nonlinearity. This nonlinearity stems from the presence of T_{back} in multiple terms, and non-linear equations typically lack straightforward algebraic solutions, especially when involving higher-order terms. Therefore, it is easier first to solve the equation in terms of T_{cavity} and then calculate the corresponding T_{back} as the following:

$$T_{cavity} = T_{out} - \frac{\left[\begin{aligned} &\alpha \cdot G_{coll} \cdot A + h_{back} \cdot A \cdot (T_{in} - T_{back}) \\ &- h_{coll} \cdot A \cdot (T_{coll} - T_{amb}) + \sigma \cdot A_{back} \cdot \epsilon_{back} \cdot (T_{in}^4 - T_{back}^4) \\ &- \sigma \cdot A_{coll} \cdot \epsilon_{coll} \cdot (T_{coll}^4 - F_{coll-sky} \cdot T_{sky}^4 - F_{coll-gnd} \cdot T_{gnd}^4) \end{aligned} \right]}{\dot{V} \cdot \rho_{air} \cdot c_{p,air}} \quad (7)$$

The equation combines several heat transfer processes such as convection (both from the backplate and exterior), radiation, and the effects of solar gain minus the losses to the ambient. In the context of dynamic insulation, the significance of T_{cavity} is twofold. Firstly, it dictates the temperature gradient across the TSC system, directly influencing the heat recovery or loss rate through the backplate.

Secondly, by quantifying the cavity temperature, the equation informs on the performance of the TSC's dynamic insulation, where the goal is to adaptively regulate heat flow to and from the building envelope. This research uses this temperature to assess the effectiveness of the TSC.

Numerous research studies have traditionally focused on realizing the heat transfer mechanisms within TSCs without giving particular attention to the phenomenon of recapturing wall heat loss. However, recent inquiries have emerged, utilizing specialized software tools, RETScreen and SWift, purposefully developed for TSC design (NRCAN, 2005; RETScreen|Natural Resources Canada, 2019; Natural Resources Canada, 2020). Both SWift and RETScreen models are rooted in the principles outlined in the International Energy Agency (IEA) report

(Carpenter and Meloche, 2002) and draw upon the insights from the Carpenter model (Carpenter et al., 2020).

While both software applications feature robust mathematical models for simulating heat transfer processes within the collector, they both employ an approximation to describe the cavity temperature of transpired solar collectors. This simplification involves defining the temperature distribution within the air cavity of the transpired solar collector as a single uniform parameter known as the effective temperature (T_{eff}). The effective temperature is estimated as a weighted average, with 2/3 derived from the collector outlet air temperature and 1/3 from the ambient temperature, as discussed in our previous work (Fawaier et al., 2021). Therefore, a new description for this important measure is needed. At the end of this section, a new definition for the backplate and the temperature distribution of the air cavity will be introduced.

Complementing this energy model, two equations merit inclusion: those representing the collector's efficiency and effectiveness. As given by the reference, these are delineated in Eqs. (8) and (9), respectively, provided by Kutscher et al. (1993a). The efficiency of TSCs is determined by the ratio of the energy gain transported by the air to the solar energy available to the collector. Thus, this ratio signifies the proportion of solar energy converted into usable heat. Meanwhile, the heat exchange effectiveness of a TSC is the actual temperature increase of the air over the potential temperature increase, given by the collector's temperature above ambient. Effectiveness values approach unity when the heat exchanger approaches ideal energy transfer efficiency, meaning it is nearly as effective at heating the air as theoretically possible given the temperature of the collector.

$$\eta_{coll} = \frac{\dot{V} \cdot \rho_{air} \cdot c_{p,air} \cdot (T_{out} - T_{amb})}{\dot{Q}_{sol}} \quad (8)$$

$$\varepsilon = \frac{T_{out} - T_{amb}}{T_{coll} - T_{amb}} \quad (9)$$

Accordingly, here in the equations T_{out} is the temperature of the air after it has been heated and is exiting the collector system to be used for heating purposes. On the other hand, T_{cavity} the air space temperature is typically immediately before entering the fan system and within the collector's cavity. In an optimal system, T_{cavity} and T_{out} should be close, indicating efficient heat transfer from the collector to the air stream. If there is a significant difference between these temperatures, it may indicate heat losses or inefficiencies in the system.

In conclusion, it can be noted that constructing a comprehensive mathematical model for the backplate temperature in transpired solar collectors is an intricate process requiring much attention to various aspects of thermal and energy balance. As a result, the solution of (7) would yield an estimation of the temperatures (T_{back} & T_{cavity}).

2.4. Mesh characteristics and solver settings

This section indicates the critical aspects of solver settings and mesh characteristics that significantly influence the accuracy and precision of the proposed numerical simulation TSC. It starts by outlining the CFD methodology's key steps, starting with system geometry and physics definition, then mesh creation, numerical solving, and results from post-processing for flow analysis.

The quality and characteristics of the mesh used in the simulation are crucial for achieving accurate results. Here, we provide an overview of the mesh employed:

- Mesh type: A structured hexahedral mesh was chosen for the simulation. This mesh type is known for accurately capturing complex geometries and flow patterns while maintaining high control over mesh quality.

- **Element size:** The mesh was designed with varying element sizes, ranging from $7\text{e-}4$ – 0.005 m , with around 300×10^3 nodes. This fine granularity allows for a detailed representation of the TSC geometry and its flow patterns.
- **Mesh quality:** To ensure the simulation's reliability, the mesh quality was carefully monitored. The skewness of elements was consistently maintained at values less than 0.01, indicating a well-structured and high-quality mesh.
- **Strategic Mesh Refinement:** Biased edge sizing was used in the TSC domain to optimize mesh resolution, as shown in Fig. 4. This technique prioritized high-resolution meshing near the walls, where the analysis concentrates on backplate temperature and heat transfer characteristics. This approach guarantees the mesh's precision in capturing critical temperature gradients and flow phenomena.

The combination of appropriate solver settings and a well-designed mesh forms the core of the simulation methodology. These choices enable the accurate representation of physical phenomena within the TSC system and ensure that the simulation results are reliable and precise.

Moreover, a mesh independence study was carried out to ensure the reliability of the simulation outcomes. The mesh was refined from an initial count of 70,000 to around 900,000 elements. It was observed that beyond 300,000 elements, the simulation results for the four key parameters (backplate temperature, airflow velocity, backplate heat flux, and heat transfer coefficient) reached stabilization. This indicates that further mesh refinement would not yield significantly different results while excessively increasing computational time. The results of the mesh independence study are illustrated in Fig. 5. It was observed that, at approximately 300,000 elements, the simulation outputs stabilized. Subsequent increases in the number of elements, up to 900,000, did not yield significant changes in these parameters.

Accordingly, these findings show that a mesh density of 300,000 elements is enough to accurately model the TSC's key physical phenomena without unnecessary computations.

The selection of appropriate solver settings is paramount to achieving reliable results in the simulation. The following configurations were employed to ensure a high-precision numerical solution:

- **Energy equation:** The energy equation was enabled within the simulation setup to account for heat transfer effects throughout the

system. This inclusion is vital for capturing the thermal behaviour of TSCs accurately.

- **Turbulence model:** The k-epsilon model with the realizable model was selected to model turbulence effects within the airflow accurately. This selection is supported by academic publications highlighting the k-epsilon turbulence model's robust representation of turbulent flow phenomena, especially in complex fluid flow scenarios, making it particularly crucial for the numerical study of TSC applications such as (Collins and Abulkhair, 2014; Tajdaran et al., 2016; Gholampour and Ameri, 2016; Croitoru et al., 2016; Li and Karava, 2012; Badache et al., 2013).
- **Wall treatment:** Enhanced wall treatment functions were applied to address boundary layer effects near solid surfaces. This ensures that the boundary conditions near the TSC surfaces are accurately represented, especially in regions with fluid-solid interactions.
- **Radiation modelling:** Radiative heat transfer within TSCs is governed by complex surface interactions. These conditions for radiative heat transfer consider the emissivity and absorptivity of the surfaces involved. These conditions are based on empirical values and material properties, allowing the model to encompass the intricate radiative phenomena. Therefore, radiative heat transfer was modelled using the surface-to-surface (S2S) radiation model. This advanced modelling approach accounts for radiative effects within the TSC system, enhancing the precision of the simulation.

2.5. Boundary conditions

The inlet boundary conditions refer to the parameters or values specified at the system's or domain's entry point. These conditions are crucial because they directly influence the system's flow. For instance, in fluid dynamics, the inlet boundary conditions may include the velocity, temperature, pressure, or concentration of a fluid entering a computational domain. These values can be measured or estimated using experimental data or a physical understanding of the problem. It is essential to ensure that the specified inlet boundary conditions are representative of the actual scenario being modelled to obtain accurate and meaningful results. This section outlines the specified boundary conditions employed to numerically solve the governing equations using Ansys Fluent, as summarised in section 2.5.

2.5.1. Inlet conditions

At the entrance region of the TSC system, the air's properties are defined. These parameters are often grounded in empirical data and experimental measurements, ensuring that the incoming air stream aligns with real-world conditions. For instance, by defining the incoming air properties such as temperature, pressure, and air stream velocity components and direction, the model replicates the dynamic nature of heat and mass transfer within TSCs.

A pressure inlet boundary condition was selected, and the total atmospheric pressure (gauge zero) aligned with standard environmental conditions. The flow direction was specified as normal to the TSC absorbance plate, mimicking practical airflow patterns encountered in practice. Moreover, a range of outdoor temperatures spanning from 0°C to 27°C , with three $^\circ\text{C}$ increments, was considered. This temperature variation allows for the thorough evaluation of TSC behaviour under different environmental conditions.

2.5.2. Outlet conditions

Conversely, the exit or outlet conditions are determined with equal precision. The temperature and velocity of the air leaving the TSC system are constrained by known values, maintaining the model's adherence to actual TSC behaviour. It is essential to ensure that these outlet conditions accurately reflect the behaviour of the air after traversing the collector, contributing to the model's overall accuracy.

The choice of an indoor temperature of 24°C was based on guidelines from both the American Society of Heating, Refrigerating and Air-

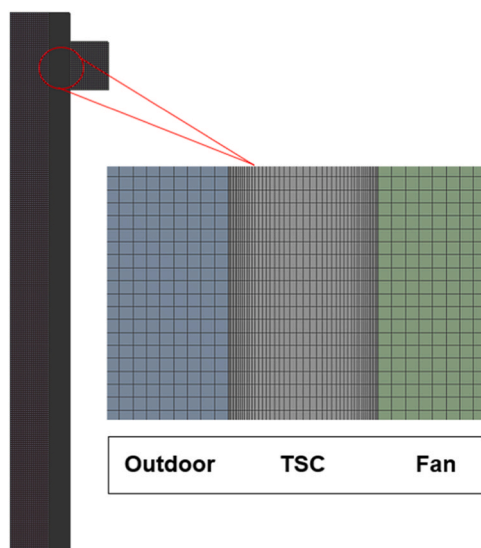


Fig. 4. Mesh characteristics of the proposed TSC numerical model showing the flow domain region (Side view).

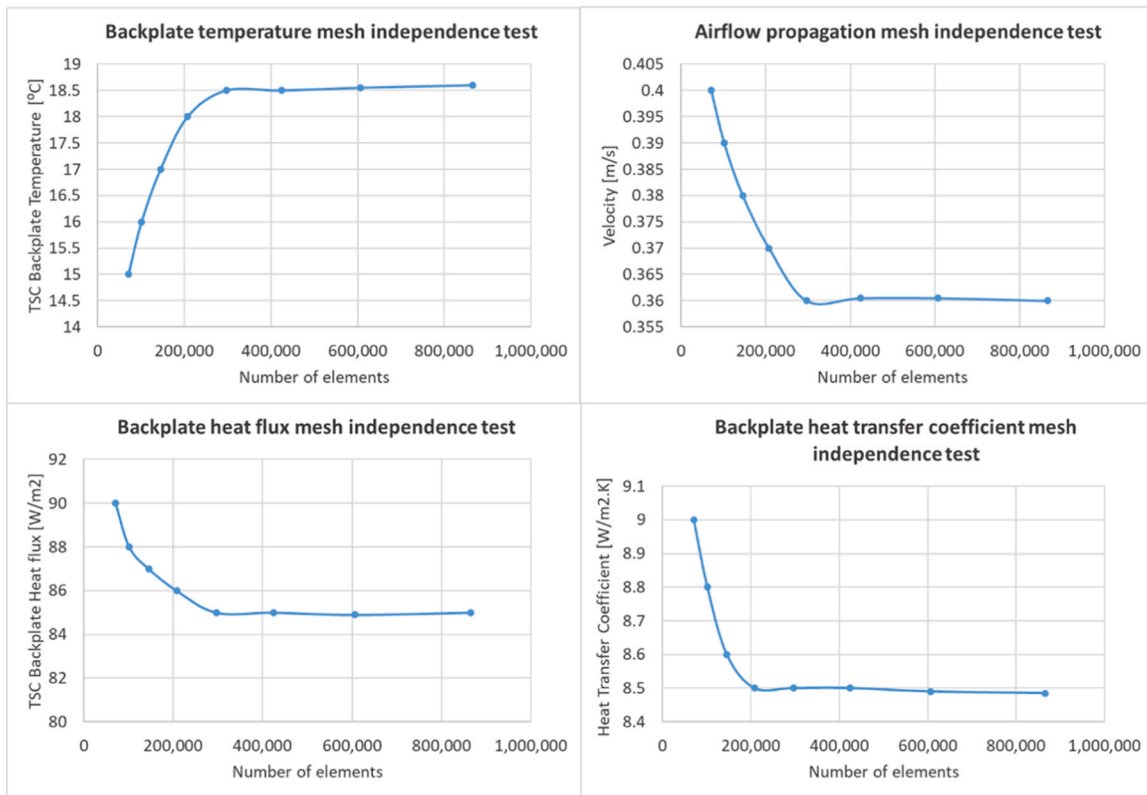


Fig. 5. Mesh independence test for backplate temperature, heat flux, heat transfer coefficient and the airflow propagation showing result stabilization at 300,000 elements.

Conditioning Engineers (ASHRAE) standard No. 55 (A. S.- ANSI/ASHRAE, 2023) and the European standard EN 15251:2007 (Wu et al., 2023), which recommends a temperature range of 20–26°C. This decision ensures occupant comfort for evaluating TSC performance in real-life conditions.

For achieving stable TSC operation, it is recommended to maintain an airflow rate ranging from 18 to 180 m³/(h·m²) (Fawaier et al., 2021). Essentially, a higher airflow rate leads to a relatively modest increase in air temperature despite the greater introduction of heat, as seen in Table 1.

A systematic approach to selecting airflow rates has been adopted to establish stable operating conditions within the TSC. A total of 10 airflow rates were examined in this study, each differing from the next by an increment of 18 [m³/(h·m²)]. This selected range intends to encompass the variety of plausible operating conditions for the TSC. By doing so, a comprehensive understanding of the TSC's performance regarding wall heat loss recapture is expected.

Following the selection of the airflow rates, the next step involves their conversion to approach velocities, a parameter often cited in existing literature and typically expressed in units of meters per second [m/s]. Essentially, the airflow rate delineates the total volume of air in circulation. In contrast, the approach velocity provides insights into the speed of this air relative to the surface of the collector. These parameters are deemed crucial for the optimal operation of transpired solar col-

lectors. This metric offers an understanding of the speed at which air traverses the collector's surface. The relationship between these two parameters is established via the equation:

$$\text{Approach velocity} = \frac{\dot{V}}{A} \quad (10)$$

Within this formula, ($\dot{V}$) denotes the airflow rate produced by the fan, and (A) represents the effective area of TSC over which the air penetrates, for instance, the area of the openings in a transpired collector.

2.5.3. Air cavity region

In addition to the outlet boundary conditions, significant focus has been placed on defining the parameters for the air cavity region within the TSC. Solar radiation and the loss coefficient for the perforated plate have been included. Solar radiation has been considered to range from 0 W/m², corresponding to night-time conditions, to 1000 W/m². This range aims to encapsulate a broad spectrum of environmental conditions, thereby providing a robust analysis.

On the other hand, the loss coefficient for the TSC's perforated plate is determined through a two-step process. Initially, the porosity of the absorber plate represented as (f) is calculated. The open area is a ratio that quantifies the proportion of the sheet occupied by holes, typically expressed as a percentage. This parameter serves as an indicator of permeability within the collector's surface. Subsequently, the perforated plate porosity is evaluated using the formula by the use of (Semenou and Rousse, 2023):

$$f = \frac{\pi}{4} \cdot \left(\frac{D^2}{P} \right) \quad (11)$$

Alternately, the other way to calculate the open area percentage of a perforated metal sheet is by using the open area percentage, which is the percentage of the total sheet area that is open or not covered by metal. It

Table 1

Recommended fan flow rate for TSC per square meter of collector area, depending on the design objective (Fawaier et al., 2021).

Design objective	Airflow rate per 1 m² of collector area
High-temperature rise	18–54 [m³/(h·m²)]
Standard operation	54–108 [m³/(h·m²)]
High air volume	108–180 [m³/(h·m²)]

is calculated using the following equation:

$$fin\% = \frac{D^2 \cdot 78.5}{C1 \cdot C2} \quad (12)$$

Where D is the hole diameter, C1 is the hole centre-to-centre distance in the horizontal direction, and C2 is the hole centre-to-centre distance in the vertical direction.

It should be noted that this equation is taken from the reference (Industrial Perforators Association, 2023), and it considers the staggered arrangement of the holes or openings in the perforated sheet in addition to the pitch (centre-to-centre distance). In this study, the following figure illustrates the holes considered inline holes with constant horizontal/vertical pitch.(Figs. 6 and 7)

Subsequently, the loss coefficient (k) is evaluated using the formula by the use of (Krotiuk, 2007):

$$k = [0.707(1 - f)^{0.375} + 1 - f]^2 \frac{1}{f^2} \quad (13)$$

This formula accounts for the resistance to airflow imparted by the perforated plate, influencing the fluid dynamics within the air cavity region. Consequently, applying the heat loss coefficient is also integral to the computation of pressure drop across the perforated plate at the TSC air cavity. The pressure drop is a critical metric for understanding the resistance encountered by airflow within the system. The equation commonly used for calculating this variable is:

$$\Delta P = 0.5 \cdot k \cdot \rho \cdot v^2 \quad (14)$$

Here, (ΔP) signifies the pressure drop, (ρ) is the air density, and (v) represents the fluid velocity. Thus, the resulting pressure drop can be embedded into the broader CFD model, enabling either empirical validation or facilitating further research studies.

2.5.4. Wall conditions

In Fluent, the wall boundary conditions for the backplate TSC were chosen to align with the study objectives. The wall is set to be stationary, which implies no motion relative to the fluid. Shear conditions are also specified, typically using the 'No Slip' condition, the standard choice for wall-fluid interactions in CFD models. The setting 'Combined External Radiation and Convection Heat Transfer' is selected for the thermal boundary conditions. This requires the definition of both the convection heat transfer coefficient and the emissivity of the wall.

On the other hand, the boundary condition for the air cavity region situated at the outer four edges forming the circumference of the TSC was established as adiabatic walls. This choice is informed by the research focus on the backplate of the collector, governed by the continuity equation. A more isolated and targeted analysis of the backplate is facilitated by assigning adiabatic conditions to these walls.

By setting these boundary conditions appropriately, the simulation will effectively capture fluid and thermal interactions at the backplate, thereby enhancing the fidelity and validity of the research outcomes. (Table 2)

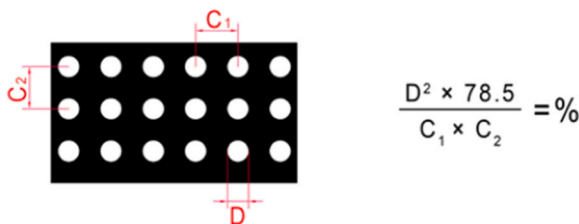


Fig. 6. Calculating open area: Round perforations and straight inline holes in perforated metal sheets.

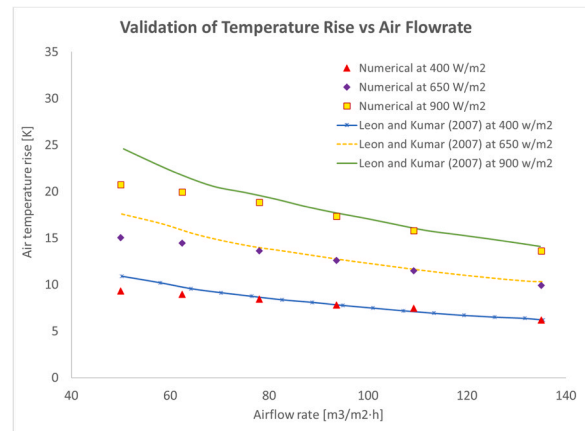


Fig. 7. Comparative analysis of air temperature rise under variable airflow between the current numerical model and Leon & Kumar's findings (Leon and Kumar, 2007).

Table 2

Main parameter and operational conditions used through the analysis.

Parameter	Value [unit]
Outdoor temperature	0–27 [°C]
Indoor temperature	24 [°C]
Operating airflow rate	18–180 [m3/h•m ²]
Solar radiation	0 – 1000 [W/m ²]
TSC surface area (WxL)	3000 × 2000 [mm]
TSC air cavity (Plenum) width	110 [mm]
Plate thickness	2 [mm]
Fan opening surface area (WxL)	1500 × 250 [mm]
Perforation (hole) diameter	1.25 [mm]
Perforation (hole) pitch	15 [mm]
Solar absorptivity	0.9 (Saidi and Hammami, 2015)
Thermal emissivity	0.85 (Saidi and Hammami, 2015)
Collector material	Mild-Steel

2.6. Model validation

In the specialised area of dynamic insulation in transpired solar collectors, the existing scientific literature is limited, posing a unique challenge for model validation. Given this context, the present study utilises a seminal reference widely acknowledged for its rigour and credibility. This source, a cornerstone in the broader field of TSCs, is a robust foundation for validating the numerical model employed.

The study's validation methodology is further strengthened by incorporating foundational principles of fluid mechanics, heat transfer, and energy conservation, as stated in section 2.3. This complementary approach aims not only to corroborate the reliability of the numerical model but also to address the complexities unique to dynamic insulation in TSC systems.

A reliable validation methodology by combining two influential and authoritative references with the specific energy balance equations. The references Leon and Kumar (2007) (Leon and Kumar, 2007) and Kutscher et al. (1993) (Kutscher et al., 1993a) have long been regarded as fundamental in the study of TSCs. This approach addresses the specificities and emerging complexities unique to dynamic insulation in TSC systems, thereby enhancing the study's reliability and academic contribution.

The key performance indicators from the Leon and Kumar study were chosen for comparative analysis by validating the temperature rise and collector efficiency against the suction airflow rate. The reference by Kutscher et al. was utilized to confirm the collector efficiency by comparing the design metrics by the literature in Table 3 with the output of the numerical model.

Table 3
Parameters for Design and Operation as Reported in Existing Literature.

	Leon and Kumar (2007)	Kutscher et al. (1993a)
Approach velocity [m/s]	0.01–0.03	0.01–0.45
Solar radiation [W/m ²]	400, 650, 900	700
Ambient temperature [°C]	30	10
Wind velocity [m/s]	1.2	0
Hole pitch [mm]	24	25
Perforation diameter (mm)	1.55	2
Porosity of absorber plate	0.33 %	0.65 %
Absorber material	Mild-steel	-
Solar absorptivity	0.95	0.9
Thermal emissivity	0.85	0.9

Existing studies on TSCs, although limited, offer invaluable data sets for such purposes. In particular, the analysis conducted by Leon and Kumar (2007) provides a detailed account of how different approach velocities influence the collector air temperature. This data set serves as an essential comparative foundation for assessing the fidelity of the current numerical model. Fig. 4 compares the rise of the collector air temperature using this reference to the results obtained through the current numerical model under varying approach velocities.

Moreover, the collector efficiency under varying approach velocities was predicted by Leon and Kumar (2007) and Kutscher et al. (1993a), comparing results obtained by both references and the numerical simulation method presented in Table 4. The maximum absolute errors of collector efficiency between the references and the current numerical model were relatively small, at less than 5 %. Consequently, it can be concluded that the numerical model accurately evaluates the collector efficiency and air temperature rise of the TSC, with results closely matching those obtained using both references.

Accordingly, Fig. 8 displays the validation of the collector efficiency of the numerical result in comparison to both references for different airflow values (Upper). The figure also shows the absolute error between the results (Lower), which ranges from low to 0.71 % and a maximum of 4.83 %.

In summary, the validation section has evaluated the numerical model’s performance by comparing its outputs with established benchmarks from credible sources in the field of TSCs. A comprehensive and reliable validation methodology has been formulated by aligning the study with seminal works and universally accepted scientific principles. The close correlation between the current model’s outputs and those from the literature underscores the robustness and reliability of the numerical simulations.

3. Results and discussion

3.1. Airflow propagation along the TSC air cavity

The propagation of airflow within the air cavity of transpired solar collectors is of considerable significance, particularly concerning the collector’s thermal efficiency and overall performance. This section delves into the complexities of airflow behaviour, elucidated through comprehensive computational fluid dynamics simulations using Ansys Fluent.

Table 4
Comparison of numerical output with results from studies by Leon & Kumar and Kutscher et al.

Approach velocity [m/s]	Collector efficiency [%]					
	Leon and Kumar (2007)	Numerical result	Error in %	Kutscher et al. (1993a)	Numerical result	Error in %
0.010	38.52	40.23	4.45	40.98	39.00	4.83
0.013	50.02	47.82	4.40	55.19	52.55	4.80
0.016	58.55	55.85	4.60	69.11	66.12	4.32
0.019	62.25	61.17	1.73	74.47	72.86	2.16
0.022	64.89	64.38	0.79	76.93	75.66	1.65
0.025	68.39	67.91	0.71	78.53	76.93	2.03

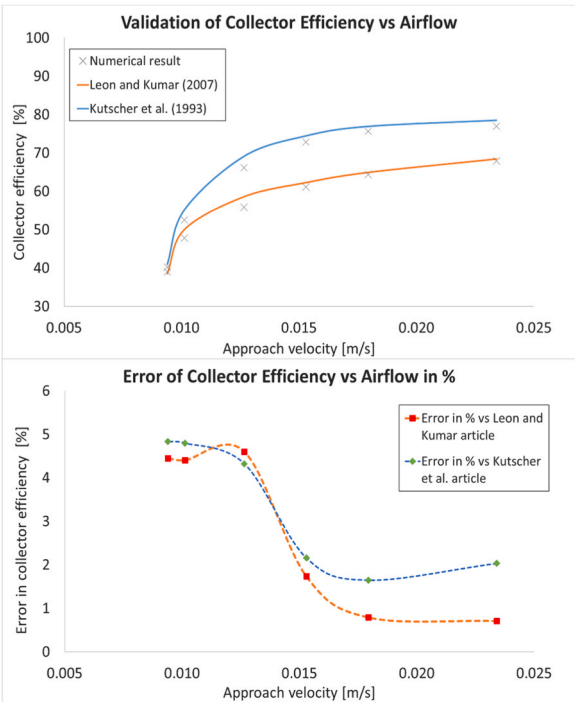


Fig. 8. Validation of collector efficiency (Upper) and absolute error (Lower) analysis for varying airflow values compared to Leon and Kumar’s (2007) and Kutscher et al. (1993a) findings.

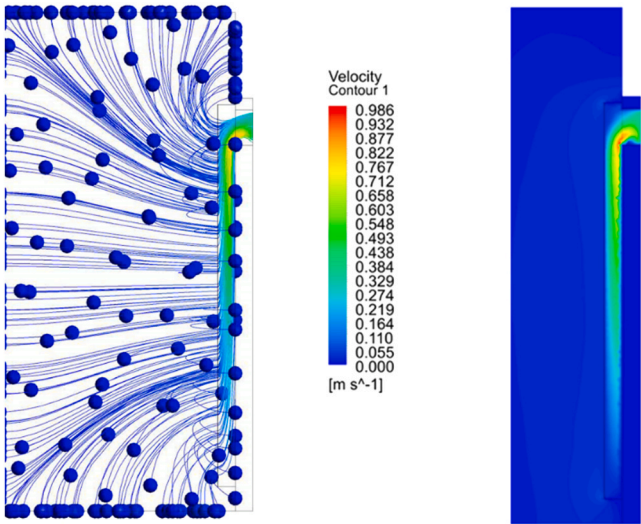


Fig. 9. Velocity contours and streamlines profile in TSC highlighting the airflow propagation build-up process (side view).

Studying the airflow propagation shown in Fig. 9 is crucial in optimising the TSC's design for maximum heat transfer while ensuring uniform air distribution throughout the collector. The airflow within the TSC air cavity, defined by its velocity profile, plays a crucial role in the collector's effectiveness in harnessing solar energy. This airflow also influences the recapturing of heat loss, as previously discussed. Therefore, as air moves through the cavity, its interaction with the backplate and the perforations influences the overall heat transfer rate and the TSC's thermal output.

The velocity contour displayed in the figure exemplifies the airflow dynamics within the air cavity of a TSC system. The gradation from deep blues to vivid reds marks the transition from low to high air velocities, capturing the uniform propagation of air throughout the cavity. This uniformity is a critical factor, as it indicates a stable and efficient airflow distribution, which is essential for understanding the TSC performance.

The dynamics of airflow within the TSC air cavity by the visualization are shown in Fig. 10. The graph presents a series of curves, each corresponding to a distinct fan flow rate, depicting the airflow's acceleration from the inlet towards the outlet.

Each curve within Fig. 10 traces the airflow velocity from the bottom of the backplate to the top. These linear trajectories indicate that the air's momentum is maintained throughout the passage, suggesting efficient airflow management within the cavity. This linear increase in velocity across the non-dimensional length is characteristic of a well-designed TSC system, where the airflow is optimized for effective heat exchange. The linear velocity profiles are critical for optimizing the TSC design. At the bottom of the backplate, the air's velocity is minimal, enhancing heat transfer due to longer contact time, and as the air moves upward, the velocity increases, potentially reducing contact time but improving convective cooling.

3.2. Heat transfer coefficient over the length

The following section delves into characterising the convective heat transfer coefficient along the length of the transpired solar collector backplate. Analyses are performed based on Ansys Fluent data visualised in 2D and 3D in Fig. 11 and Fig. 12. These exhibit the heat transfer coefficient influenced by various airflow rates across the non-dimensional backplate length.

The heat transfer coefficient plays a vital role in the thermal dynamics of TSCs, signifying the heat exchange efficiency between the backplate and the air stream inside the collector. Fig. 11 illustrates how this coefficient changes concerning the backplate's position, showing its variation with distance from the air inlet. A prominent observation is an increase in heat exchange effectiveness with non-dimensional length, indicating an improvement in heat transfer as the air progresses along the backplate.

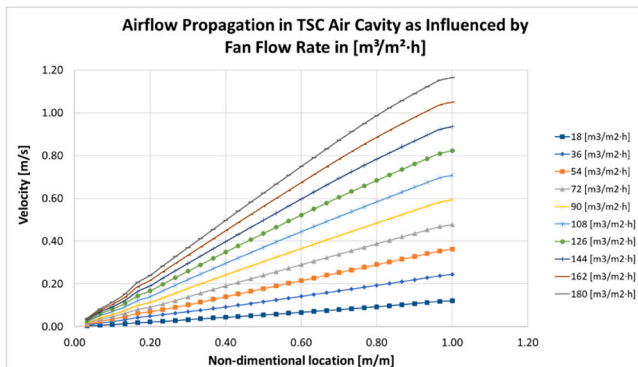


Fig. 10. Airflow velocity profiles within the TSC cavity at varying fan flow rates.

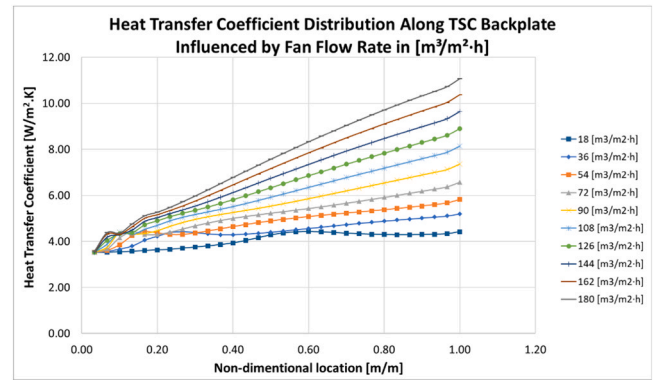


Fig. 11. Heat transfer coefficient variation on the TSC backplate as a function of fan airflow.

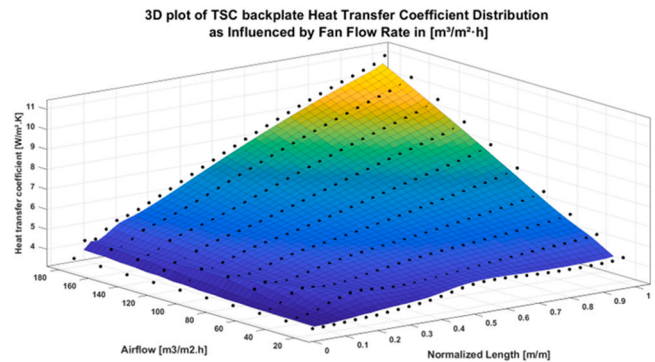


Fig. 12. Heat transfer coefficient variation on the TSC backplate as a function of fan airflow in the 3D surface plot.

The observed trends suggest a pronounced peak in the heat transfer coefficient at the mid-high positions of the TSC backplate when subjected to lower airflow rates. This peak indicates an enhanced convective heat transfer efficiency resulting from the gentler air movement, allowing for a more stable and gradual heat transfer due to buoyancy and less boundary layer disturbance.

As the airflow rates are elevated, this distinct peak begins to diminish and transitions towards the lower edge of the backplate. The increased flow introduces turbulence, which disrupts the boundary layer, fostering enhanced mixing and leading to a more uniform distribution of the heat transfer coefficient across the backplate. The impact of this turbulence is twofold: it diminishes the distinct peak observed at lower flow rates. It promotes a more uniform thermal profile across the entire surface area of the backplate.

Lower flow rates would be beneficial for scenarios requiring intensified heat transfer at the top of the backplate. Conversely, higher flow rates would be advantageous to achieve a uniform thermal distribution across the backplate, especially in conditions that might otherwise lead to overheating or in situations where a consistent temperature profile is desired.

3.3. Back plate temperature over the length

This section explores the results and discussion of the numerical simulation study on the back plate temperature of transpired solar collectors conducted using Ansys Fluent. The study's primary focus is understanding how the temperature distribution along the length of the back plate varies under different operational and environmental conditions.

The simulation results have been systematically studied over a significant range of ambient temperatures, fan flow rates, and solar

radiation intensities. The ambient temperatures examined span from 0°C to 27°C, fan flow rates from 18 to 180 cubic meters per hour, and solar radiation intensities from 0 to 1000 W/m². The comprehensive data collection at 30 normalized points provides a high-resolution picture of the temperature distribution.

The investigation started by showing the temperature contour in Fig. 13, which outlines the complex thermal dynamics on the TSC backplate. The contour illustrates a non-linear temperature profile that shifts from cooler blues to warmer reds, clearly visualising the heat transfer mechanisms. This gradient is a direct manifestation of the interplay between solar radiation absorption, the cooling effect generated by the fan's airflow, and the thermal properties of the backplate material. Such localized variations in temperature underscore the nuanced interaction between the TSC's design parameters and environmental influences, showcasing the system's thermal response to variable operational conditions.

3.3.1. Night-time backplate temperature investigation

This section investigates the nocturnal thermal characteristics of the TSC backplate. Here, the absence of solar irradiance results in a temperature profile primarily influenced by the ambient thermal conditions and the rate of the working fan. An analysis of the TSC's thermal performance during nocturnal hours is presented in Fig. 14, where distinct airflow regimes are correlated with uniform temperature distributions across the backplate.

The comprehensive analysis of the TSC backplate temperature distribution during nocturnal conditions yields five distinct insights:

1. **Thermal Equilibration Time:** At lower airflow rates, the backplate temperatures are higher due to the prolonged period available for heat absorption from the recaptured indoor heat loss. This suggests that the backplate reaches a thermal equilibrium over a longer duration, allowing for greater thermal uptake and higher surface temperatures.
2. **Thermal Entry Length:** The data suggests a pronounced thermal entry length at lower flow rates, characterized by a noticeable temperature gradient indicative of the initial heat transfer phase. Higher flow rates seem to shorten this thermal entry length, likely due to increased convective forces that enhance the heat transfer rate, leading to a more uniform temperature distribution.
3. **Uniformity at Elevated Flow Rates:** The temperature profiles at higher airflow rates display consistency in temperature distribution, indicative of an efficient convective heat transfer mechanism that minimizes temperature differentials along the backplate.

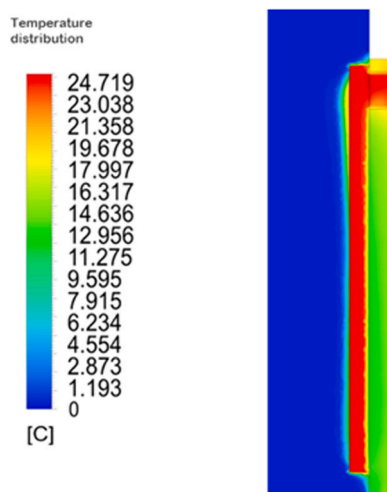


Fig. 13. Temperature contour of the TSC highlighting the gradient from cooler to warmer regions due to environmental and operational influences (side view).

4. **Ambient Temperature Impact:** The ambient temperature is a critical determinant of the TSC backplate temperature profiles. Higher ambient temperatures result in higher backplate temperatures, likely due to a smaller thermal gradient between indoor and outdoor environments, reducing heat loss's driving force.
5. **Flow Rate Management for Optimal Performance:** These insights underline the importance of managing the flow rate to optimize the TSC's performance. Lower flow rates are beneficial for enhancing heat gain during cooler ambient temperatures. In contrast, higher flow rates are preferable for maintaining temperature uniformity across the backplate during warmer conditions.

The influence of ambient temperatures and fan flow rates on the TSC backplate's nocturnal thermal distribution is depicted in Fig. 15, where the thermal entry length varies noticeably across different cases.

The figure illustrates three distinct temperature profiles along the non-dimensional location of the TSC backplate for night-time scenarios, each likely corresponding to different ambient temperature and fan flow rate conditions. The profiles exhibit varying thermal entry lengths, where the temperature gradient is most pronounced. The profile labelled "Case 1" remains relatively flat, suggesting a stable temperature across the backplate, possibly due to low fan flow rates or high ambient temperatures. "Case 2" shows a notable temperature drop-off before stabilizing, which could indicate higher fan flow rates or lower ambient temperatures affecting the backplate" thermal behaviour. "Case 3" demonstrates an even steeper initial gradient, implying a significant influence of thermal entry length, and then plateaus similar to "Case 2".

These insights have implications for the operational strategy of TSCs. Controlling airflow rates becomes critical to optimize the system for varying ambient conditions and desired thermal characteristics. Understanding the interplay between fan flow rates, ambient temperature, and the resultant thermal behaviour of the TSC backplate is essential for improving the system's thermal performance and energy efficiency.

3.3.2. Daytime backplate temperature investigation

In contrast, the daytime scenario illustrated by Fig. 16 exhibits a marked increase in temperature. The simulation data reveal the expected rise in temperature due to solar irradiance and a complex interaction between the rate of solar heat gain and the backplate's thermal dynamics. Observations from the simulations emphasise the TSC's sensitivity to operational and environmental conditions and underscore the necessity for a design that accommodates a wide spectrum of temperatures and solar exposure levels.

The new set of graphs describes the TSC backplate temperature profiles during daylight conditions with varying solar radiation intensities, ranging from 200 to 1000 W/m², as follows:

1. **Solar radiation intensity and temperature increase:** An increase in solar radiation from 200 to 1000 [W/m²] results in a marked escalation in temperature across the TSC backplate. This trend aligns with the expectation that a more significant solar energy influx leads to increased heat absorption by the backplate.
2. **Non-linear temperature gradient:** The lower section of the TSC backplate, specifically in the region between the non-dimensional locations of 0 and 0.3, demonstrates a pronounced non-linear temperature gradient. This steep increase in temperature can be attributed to a higher heat flux in this area. Factors contributing to this condition include potentially higher rates of heat recapture from indoor spaces and reduced airflow at this lower portion of the TSC, leading to less convective heat removal and higher temperatures.
3. **Uniformity with limitations:** Beyond the initial rapid temperature rise at the lower section, the TSC backplate's temperature profile levels off, indicating a uniform heat distribution. This points to an efficient thermal design, although the tendency of the temperature to plateau suggests that the backplate's capacity to absorb heat is not infinite and could be a limiting factor in the system's performance.

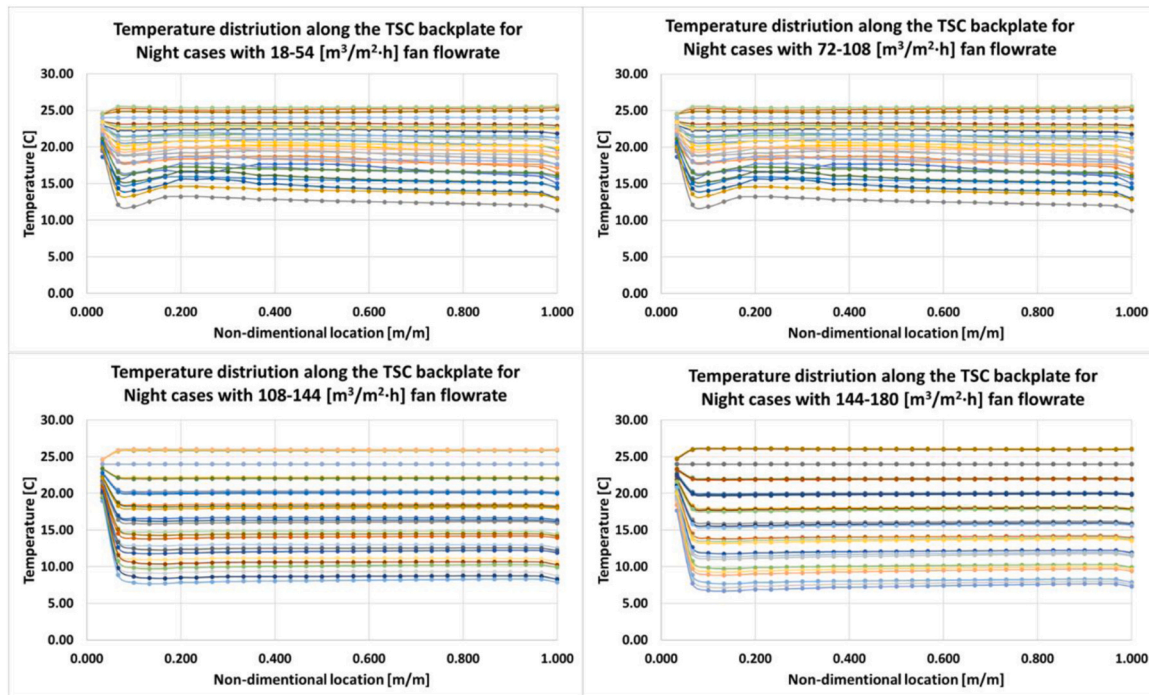


Fig. 14. Night-time temperature profiles for various fan flow rates across the TSC backplate.

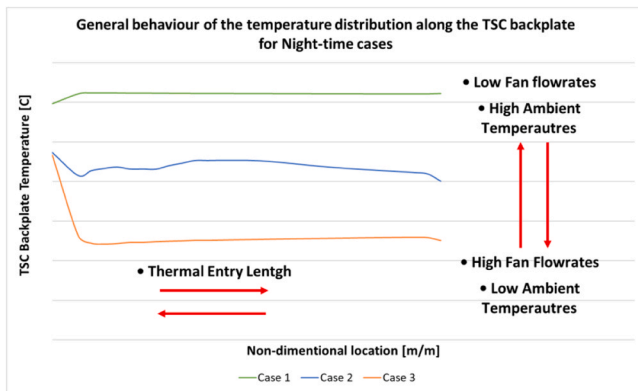


Fig. 15. Comparative temperature profiles of the TSC backplate during night-time under varying ambient and flow conditions.

4. The interplay of ambient temperature and fan flow rate: The ambient and fan flow rates affect the temperature profiles. Higher backplate temperatures can indicate higher ambient temperatures combined with slower fan flow rates, allowing increased heat absorption from the backplate. On the contrary, lower temperature profiles may be due to cooler ambient conditions and faster fan flow rates, enhancing convective heat dissipation from the backplate.
5. Dynamic adjustment of operational parameters: Optimal thermal performance of the TSC necessitates the dynamic adjustment of the fan flow rate according to fluctuations in ambient temperature. This dynamic response is crucial to maintain the desired temperature profiles on the backplate and to mitigate the risk of overheating or insufficient thermal utilization during operation.

Together, these points offer a comprehensive analysis of the thermal dynamics of TSCs under varying solar radiation conditions, emphasizing the importance of considering design and operational strategies to optimize performance and prevent thermal inefficiencies.

3.4. Backplate heat flux over the length

This section studies the longitudinal heat flux variation along the TSC backplate, a critical parameter revealing the system's thermal performance and energy transfer efficiency. This section, and after the term heat flux, will represent the backplate heat transfer recaptured by the mean of TSC. Employing Ansys Fluent for numerical simulations, the study produces a detailed heat flux investigation.

The analysis will then be divided into two distinct operational periods: night-time and daytime. In this case, this section covers how heat flux varies along the length of the backplate, influenced by both the design parameters of the TSC and the operational conditions under which it functions.

3.4.1. Night-time backplate heat flux investigation

The nocturnal analysis reveals a significant correlation between airflow rate and heat flux along the backplate. As the airflow rate increases, a corresponding rise in heat flux is observed, indicating an efficient heat exchange process that is effectively captured. The impact of fan flow rates on the distribution of heat flux along the TSC backplate during night-time conditions is depicted in Fig. 17, revealing the relation between airflow and thermal performance.

In light of the provided notes, here's a refined analysis that closely aligns with the night-time operational characteristics of TSCs:

1. Influence of fan flow rate on heat recapture: At a constant ambient temperature, higher fan flow rates indicate a more effective recapture of lost heat from the indoor space. This could be due to increased air movement facilitating heat transfer from the backplate to the indoor environment. Conversely, at lower airflow rates, the heat flux distribution becomes more uniform and less intense, suggesting that the air has less capacity to recapture the heat loss from the indoor space due to reduced convective force.
2. Heat flux profile and air flow rate: The heat flux profiles tend to flatten at lower airflow rates, which denotes a more uniform heat distribution along the TSC backplate. As the airflow rate increases, a steeper gradient in the heat flux profile becomes apparent, likely due

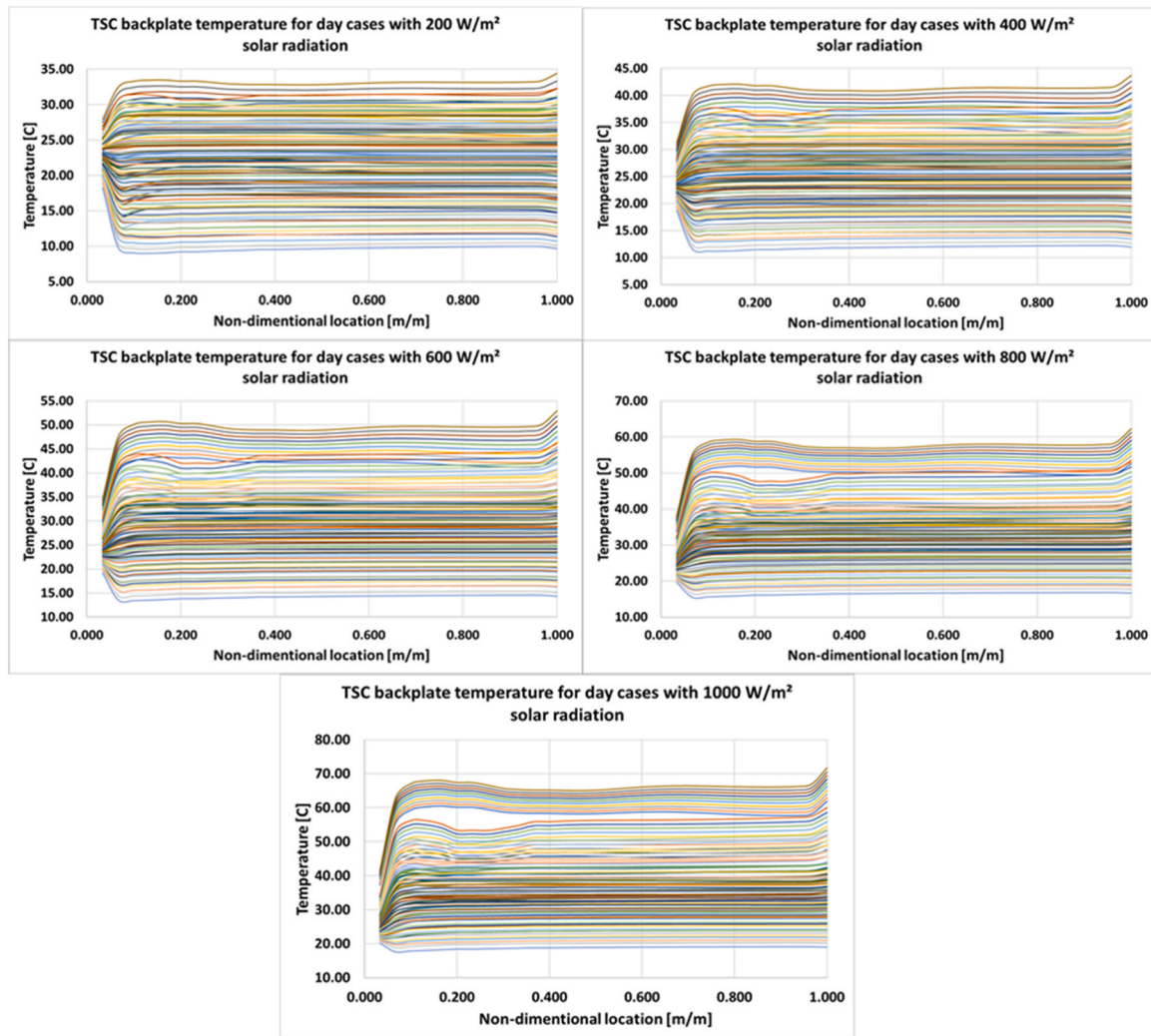


Fig. 16. Daytime TSC backplate temperature profiles under escalating levels of solar radiation.

- to the enhanced convective transfer effects that redistribute the heat along the backplate more rapidly.
3. Localized heat transfer variations: The graphs still indicate variations in heat flux values along the TSC backplate. These variations may now be understood as the result of localized thermal dynamics, which include the interplay of ambient temperature and the varying efficiency of convective heat recapture at different points along the backplate.
 4. Interpretation of negative heat flux: Negative heat flux values at night signify that the TSC backplate transfers heat to the indoor environment. This phenomenon is observed when the ambient temperature is at or above the indoor temperature, prompting the backplate to act as a heat source due to its retained warmth from daytime operation.
 5. Strategies for night-time optimization: The data suggests fine-tuning the fan flow rates is essential for night-time TSC optimization. By calibrating the fan speeds with the ambient temperature, one can control the heat flux to ensure efficient indoor heat recapture when ambient temperatures are lower and prevent excess heat loss or gain when ambient temperatures are higher.

Moreover, the comprehensive depiction of heat flux profiles in Fig. 18 highlights the critical influence of fan flow rates and ambient temperature gradients on the TSC backplate's thermal dynamics throughout the nocturnal phase.

The figure displays four night-time scenarios of heat flux distribution along the TSC backplate, distinguished by varying fan flow rates and ambient temperatures. The depicted lines reveal how these factors influence heat flux behaviour. Scenarios with higher fan flow rates correspond to greater heat flux, suggesting enhanced convective heat transfer. In contrast, scenarios with lower fan flow rates show reduced heat flux, potentially indicating decreased convective cooling and possibly higher temperatures on the backplate. The annotations compare indoor and outdoor temperatures, highlighting the temperature differential as a critical driver of heat transfer. Scenarios, where outdoor temperatures are lower than indoor ones, show a heat flux pattern characteristic of heat loss to the surroundings that the TSC can recapture.

These insights are essential in grasping the thermal dynamics of TSCs during night-time, providing critical information to improve their performance. This knowledge ensures that the system functions efficiently under various nocturnal conditions.

3.4.2. Daytime backplate heat flux investigation

As the investigation examines the TSC backplate heat flux during daytime, the impact of solar radiation on its thermal performance becomes apparent. This section, grounded in numerical simulations, examines the backplate's capacity to absorb and distribute solar energy, considering the variation in its length. Central to this analysis is a series of plots in Fig. 19, each representing different solar irradiance levels.

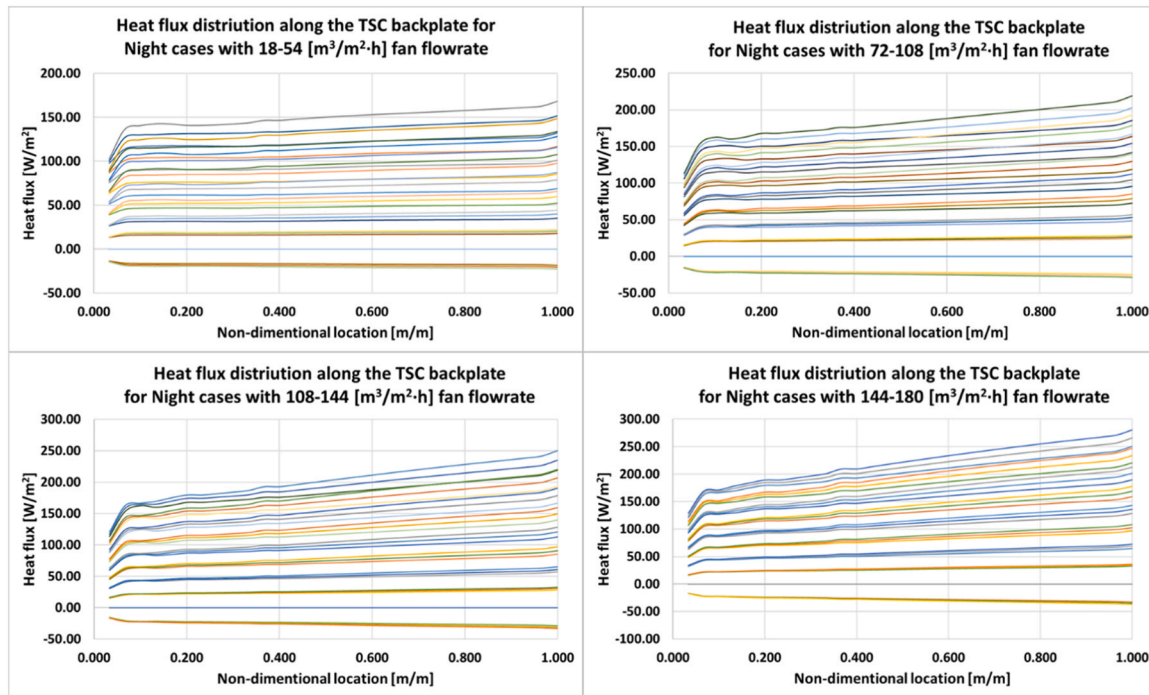


Fig. 17. Night-time heat flux profiles across the TSC backplate at varying fan flow rates.

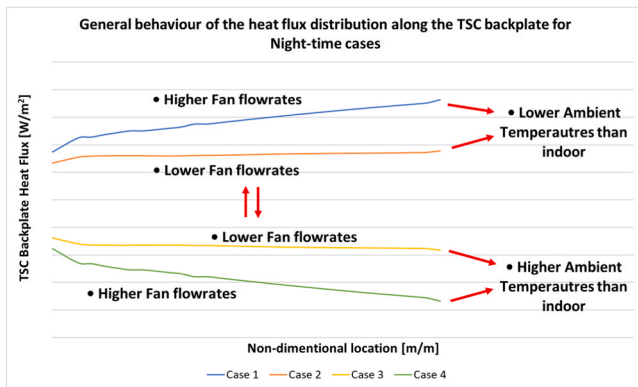


Fig. 18. Heat flux profiles at varying nocturnal fan flow rates and ambient temperature gradients across the TSC backplate for the night-time cases.

These plots are pivotal in decoding the heat transfer processes within the TSC. They play a key role in pinpointing the balance between energy absorption and loss, which is crucial for designing and optimising TSC systems for improved energy efficiency.

Examining the TSC backplate heat flux for daytime cases with varying solar radiation intensities from 200 to 1000 W/m², the following observations can be noted:

1. **Scaling heat flux with solar intensity:** An examination of the Y-axis values across the series of graphs reveals that the span of heat flux readings adjusts with the solar radiation levels. Specifically, at lower to medium solar radiation intensities of 200–600 W/m², the heat flux predominantly shows positive values with a greater slope, reflecting significant wall heat loss recapture from the interior. This indicates that the TSC backplate effectively reclaims the thermal energy escaping from the building. Conversely, the emergence of negative heat flux values becomes more prevalent as the solar radiation intensity reaches higher levels of 800–1000 W/m², suggesting that the backplate starts to absorb heat from the external environment, which

can be attributed to the intense solar heat exceeding the heat recovery capabilities of the TSC system.

2. **Heat flux profile characteristics:** The profiles across different solar radiation intensities reveal a nuanced behaviour. For instance, at mid-range solar intensities of 400–600 W/m², the profiles show a marked increase in heat flux towards the middle of the backplate, which could be due to the optimal absorption zone of the TSC. Conversely, at the highest intensity of 1000 W/m², the profiles indicate potential thermal saturation with a flatter trend, suggesting a limit to the heat absorption capacity of the backplate material or a maximum efficiency threshold of the TSC design.
3. **The transition from positive to negative heat flux:** The transition from positive to negative heat flux within the same solar radiation level is an interesting observation that needs to be studied in depth. Under high solar radiation, the backplate experiences significant heat gain. Still, the fan flow rate and ambient temperature can drastically alter this balance. At high fan flow rates and low ambient temperatures, the TSC is more efficient at shedding excess heat, reflected by sustained positive heat flux values. However, as the fan flow rate decreases and ambient temperatures rise, the backplate may reach thermal equilibrium with its surroundings or even absorb heat from the ambient air, leading to negative heat flux readings.
4. **Influence of Non-Dimensional Location:** The non-dimensional location along the backplate correlates with varying heat flux values, hinting at a complex interplay of thermal gradients. At locations closer to the TSC lower part (near 0 [m/m]), higher heat flux values may decrease towards the fan inlet (near 1 [m/m]). This could reflect the heat capture process as the absorbed solar energy is gradually transferred to the air flowing over the backplate.
5. **Implications for TSC daytime operation:** The insights from the heat flux distribution underscore the necessity for an intelligent operational strategy. Such a strategy would incorporate real-time adjustments of fan flow rates and possibly active or passive thermal management techniques to regulate the heat flux across the backplate, thereby optimizing the TSC's performance throughout the varying conditions of the day.

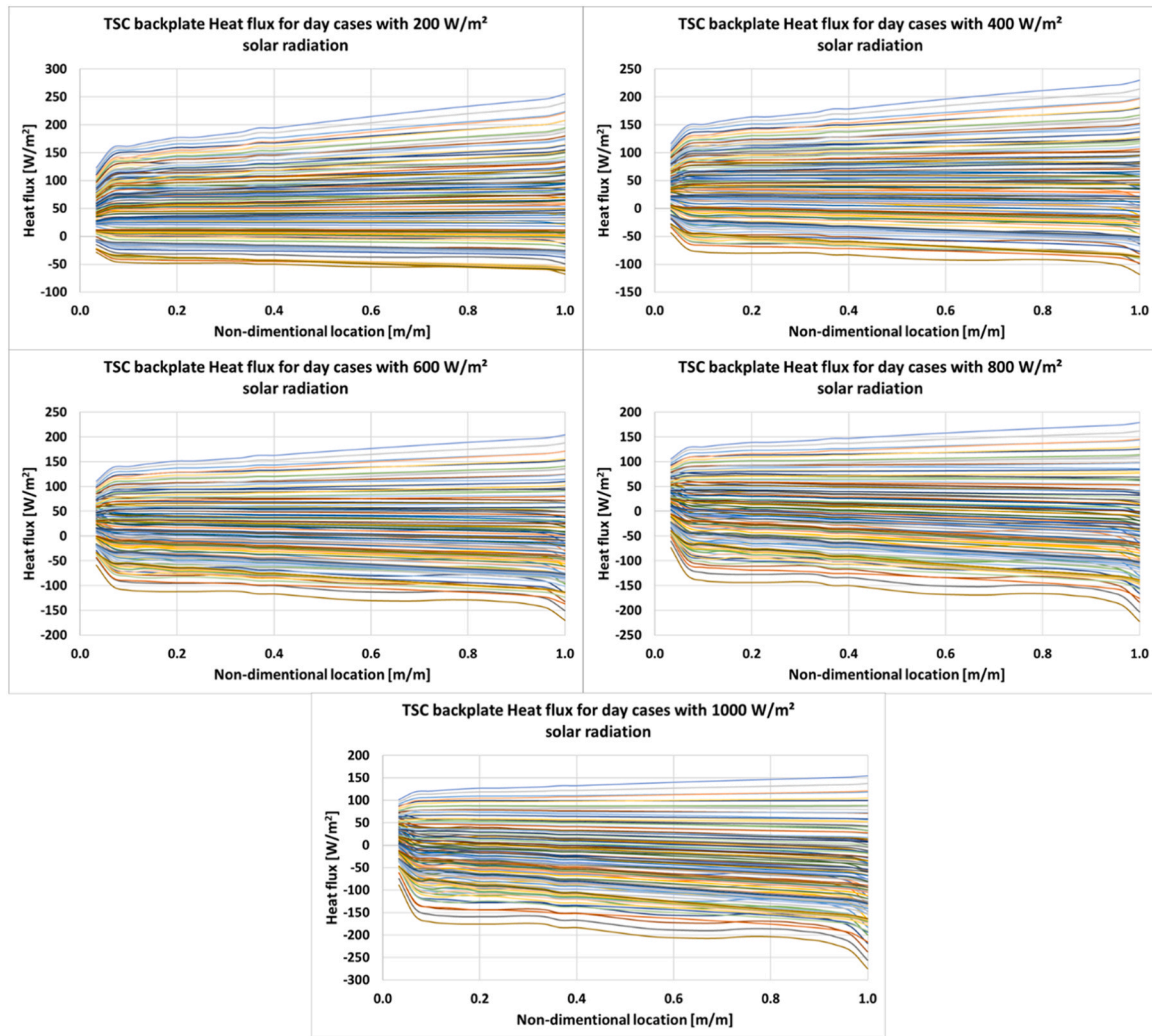


Fig. 19. Daytime heat flux variation on the TSC backplate across a range of solar radiation intensities.

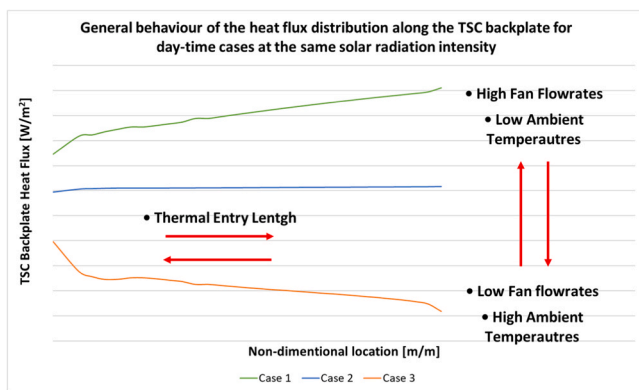


Fig. 20. Comparative analysis of heat flux along the TSC backplate under daylight conditions, showing the influence of fan flow rates and ambient temperatures across three distinct cases.

Finally, Fig. 20 illustrates the daytime heat flux patterns on the TSC backplate under a constant solar irradiance level. It highlights the most notable cases arising from varying fan flow rates and fluctuating ambient temperatures. Case 1 (green) depicts a consistent rise in heat flux due to higher ambient temperatures and fan flow rates, optimizing the balance between heat absorption and dissipation. Case 2 (blue)

represents a distinctive case where the fan flow rate is reduced while ambient temperature increases. This specific tuning results in a semi-constant heat flux pattern along the backplate's length. Meanwhile, Case 3 (orange) starts with a heightened heat flux of decreasing pattern along the backplate, reflecting lower fan flow rates paired with warmer ambient conditions. This nuanced analysis of heat flux under varying conditions is crucial for refining TSC system design and operation, aiming to achieve optimal thermal efficiency in diverse solar conditions.

3.4.3. Effect of solar radiation on dynamic heat loss recapture

This section investigates the effects of solar radiation on heat loss recapture in transpired solar collectors. Two distinct investigative processes were engaged to evaluate the influence of solar radiation on transmission heat loss and its subsequent recapture by the TSCs. These analyses incorporated a range of airflow rates from 18 to 108 m³/m²·h and ambient temperatures from 0 to 27°C, as depicted in the forthcoming Fig. 21.

The initial phase of the study focused on establishing the upper bounds of transmission heat loss that the TSC could effectively recapture under nocturnal conditions. This phase involved detailed modelling to quantify the maximum transmission losses that could be recaptured by the TSC, providing a baseline against which to measure the impact of solar radiation.

Subsequently, the second phase exposed the TSCs to a gradient of solar radiation intensities from 200 to 1000 W/m² at the adiabatic

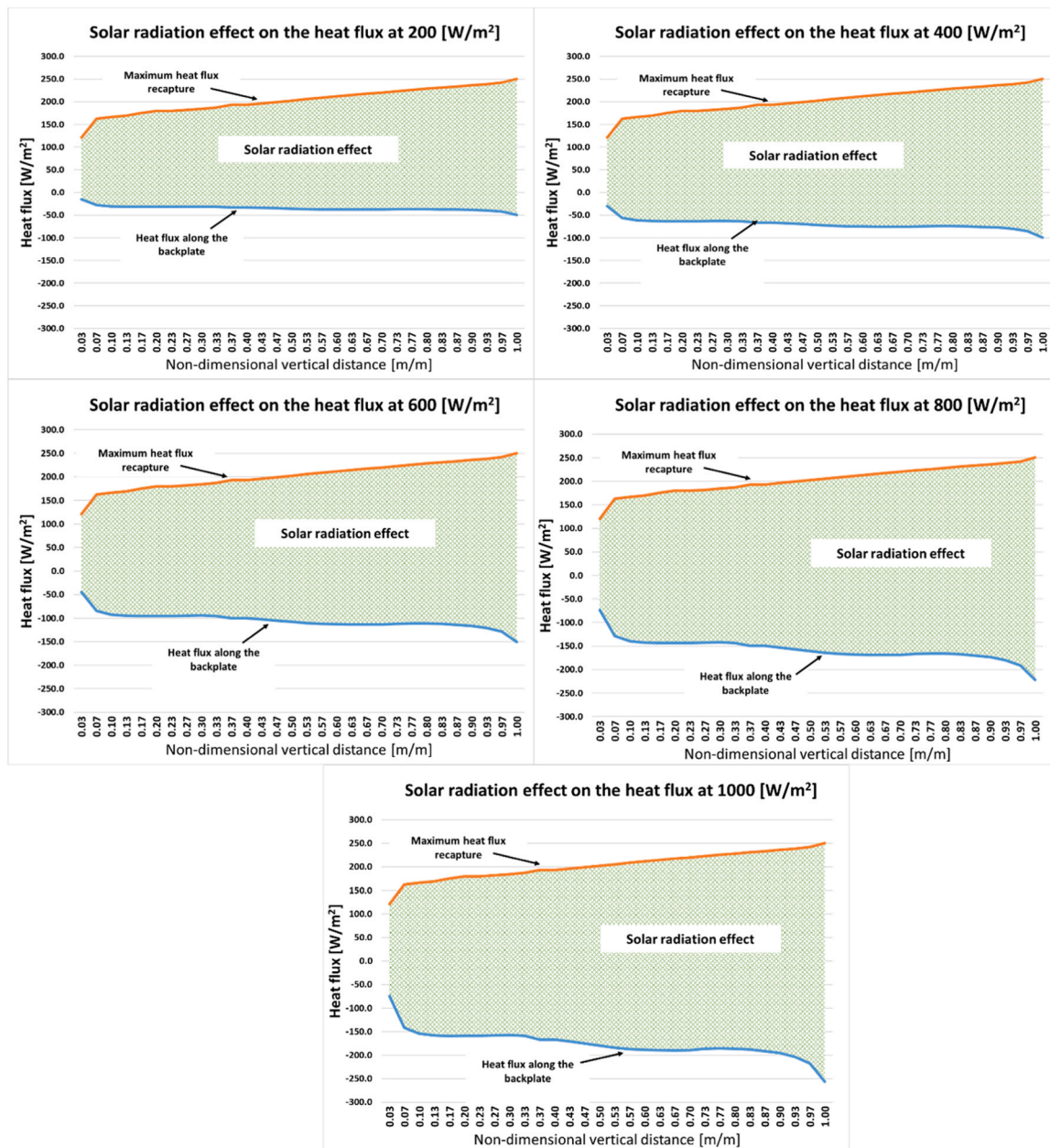


Fig. 21. Dynamic heat flux recapture profiles under varying solar radiation intensities from 200 to 1000 W/m^2 .

collector backplate. This phase was critical in assessing the system's response under extreme conditions where no recapture is allowed and only the effect of the solar radiation exists. This would give the backplate the perfect conditions for reaching the most elevated temperatures and developing a negative heat flux. A comparative analysis was conducted by measuring the heat flux recaptured along the backplate and juxtaposing it with the maximum transmission losses established in the first phase.

The data revealed a marked mean reduction in wall heat loss recapture attributed to solar influence, quantified as follows for the respective solar radiation levels:

- At 200 W/m^2 , there was a 125 % reduction in heat loss recapture.
- At 400 W/m^2 , the reduction increased to 147 %.
- At 600 W/m^2 , a further increase to 172 % was observed.
- At 800 W/m^2 , the decline reached 203 %.
- Finally, at 1000 W/m^2 , the reduction peaked at 241 %.

These findings underscore the significant impact that solar radiation has on TSC performance. The escalating percentage reductions in heat loss recapture with rising solar radiation levels indicate that the TSC's ability to reclaim transmission losses diminishes as solar intensity increases. This inverse relationship is a pivotal consideration in the design and operational management of TSC systems, particularly in regions with high solar irradiance.

4. Conclusions

This study conducts an extensive numerical investigation into the thermal dynamics of transpired solar collectors (TSCs) using Ansys Fluent, particularly emphasising the temperature distribution, heat transfer characteristics, and airflow propagation within the TSC backplate. The research employs a robust CFD model with selected solver settings, turbulence models, and radiation modelling approaches to capture the intricate physical phenomena within TSCs accurately. The numerical model was validated against established benchmarks and

credible sources in the TSC field, ensuring the reliability and accuracy of the simulations.

The investigation encompasses a comprehensive range of operational and environmental conditions, including ambient temperatures from 0°C to 27°C, fan airflow rates from 18 to 180 [m³/m².h], and solar radiation intensities from 0 to 1000 [W/m²]. This broad spectrum of parameters provides a detailed understanding of the backplate's thermal behaviour and airflow dynamics under diverse scenarios. The key findings and achievements of this research are summarized as follows:

■ Airflow Propagation and Velocity Profile:

- The airflow within the TSC air cavity exhibited a uniform propagation and linear velocity increase from the inlet towards the outlet, indicating efficient airflow management and momentum maintenance throughout the passage.
- The linear velocity profiles across the non-dimensional length of the backplate are characteristic of a well-designed TSC system optimized for effective heat exchange.

■ Backplate Heat Transfer Coefficient:

- The heat transfer coefficient along the TSC backplate exhibited a pronounced peak at mid-high positions under lower airflow rates, indicating enhanced convective heat transfer efficiency due to gentler air movement and reduced boundary layer disturbance.
- As airflow rates increased, the distinct peak diminished, and the heat transfer coefficient distribution became more uniform across the backplate, facilitated by increased turbulence and boundary layer disruption.
- This observed behaviour suggests that operational parameters can be fine-tuned to achieve specific thermal performance goals, with lower flow rates beneficial for intensified heat transfer at the top of the backplate and higher flow rates advantageous for uniform thermal distribution across the entire surface.

■ Backplate temperature distribution shows the following key characteristics:

A) Nighttime temperature distribution

- A non-linear temperature profile was observed on the TSC backplate, transitioning from cooler to warmer regions influenced by the operational and environmental properties.
- Lower airflow rates resulted in higher backplate temperatures, longer thermal equilibration times, and increased heat absorption, indicating enhanced thermal uptake.
- Higher airflow rates led to a more uniform temperature distribution across the backplate, suggesting efficient convective heat transfer.

B) Daytime temperature distribution

- The backplate temperature increased proportionally with rising solar radiation intensities from 200 to 1000 W/m².
- A steep temperature increase occurred in the lower section of the TSC backplate (between non-dimensional locations of 0 and 0.3 m/m), attributed to higher heat flux due to potentially higher indoor heat recapture rates and reduced airflow.

■ Backplate heat flux distribution shows the following key characteristics:

A) Nighttime heat flux analysis:

- A correlation between airflow rate and heat flux along the TSC backplate indicated an efficient heat exchange process. Increasing

airflow rates resulted in higher heat flux, suggesting a more effective recapture of indoor heat losses.

- At lower airflow rates, the heat flux distribution became more uniform and less intense, indicating reduced indoor heat loss recapture capacity due to diminished convective force.
- Negative heat flux values were observed at night, implying the TSC backplate transferred heat to the indoor environment when the ambient temperature was at or above the indoor temperature, highlighting the TSC's potential for passive heating.

B) Daytime heat flux analysis:

- Under high solar radiation, while the backplate experienced significant heat gain, the fan flow rate and ambient temperature drastically altered the balance between this gain and heat dissipation.
- The TSC demonstrated the ability to effectively shed excess heat under certain conditions, as reflected in sustained positive heat flux values.

■ Effect of solar radiation on dynamic heat loss recapture

- Solar radiation significantly impacted the TSC's ability to recapture transmission heat losses.
- An inverse relationship was observed between solar radiation intensity and heat loss recapture, with reductions ranging from 125 % at 200 W/m² to 241 % at 1000 W/m².
- This finding underscores the necessity for intelligent operational strategies involving real-time adjustments of fan flow rates and potential active or passive thermal management techniques to regulate heat flux across the backplate, optimizing TSC performance under varying solar conditions.

The research findings provide valuable insights for optimizing TSC design and operation, contributing to sustainable and energy-efficient building solutions. However, further research is necessary to assess their long-term performance, effectiveness as dynamic insulation, efficiency across various climates, maintenance requirements, and cost-effectiveness. Future research should also explore integrating these systems with advanced controls and energy storage and examine synergies with sustainable practices to enhance building energy management.

CRediT authorship contribution statement

Mohammad Fawaier: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Goda Róbert:** Writing – review & editing, Visualization, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Balázs Bokor:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

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References

- A. S.- ANSI/ASHRAE, undefined 55, and undefined 1992, Thermal Environmental Conditions for Human Occupancy, *cir.nii.ac.jp*, Accessed: Sep. 28, 2023. [Online]. Available: (<https://cir.nii.ac.jp/crid/1574231875437564672>).
- Appiah, M.K., Gyening, E.K., Teye, P.K., Frimpong, C., Nsowah, A., 2023. The implications of energy literacy on energy savings behavior: a model of contingent effects of energy value and attitude. *Energy Rep.* vol. 10, 72–85. <https://doi.org/10.1016/J.EGYR.2023.06.008>.
- Arulanandam, S.J., Hollands, K.G.T., Brundrett, E., 1999. A CFD heat transfer analysis of the transpired solar collector under no-wind conditions. *Sol. Energy* vol. 67 (1–3), 93–100. [https://doi.org/10.1016/S0038-092X\(00\)00042-6](https://doi.org/10.1016/S0038-092X(00)00042-6).
- Badache, M., Rousse, D.R., Hallé, S., Quesada, G., 2013. Experimental and numerical simulation of a two-dimensional unglazed transpired solar air collector. *Sol. Energy* vol. 93, 209–219. <https://doi.org/10.1016/J.SOLENER.2013.02.036>.
- Balázs, B., Hacer, A., Dogan, E., László, K., 2019. View of the potential of solar air heating in the turkish industrial sector. *Period. Polytech. Mech. Eng.* vol. 63 (1), 57–66. (<https://pp.bme.hu/me/article/view/13028/8182>). Accessed: Sep. 06, 2020. [Online]. Available.
- Barker, G., Kiatreungwattana, K., 1997. Transpired solar collector systems: a major advance in solar heating. (Jun.)*ASME 2011 5th Int. Conf. Energy Sustain. ES 2011*, no. PARTS A, B, C. 113–118. 10.1115/ES2011-54256..
- Cao, S., Hollands, K.G.T., Brundrett, E., 1993. Heat exchange effectiveness of unglazed transpired-plate solar collector in 2D flow. *Proc. ISES Sol. World Congr.* 361–366.
- S. Carpenter, S. Daniels, S. Kemp, J. Kokko, and G. Van Decker, New Tools for Assessing the Performance of Solar Ventilation Air Heating Systems (Miscellaneous) | ETDEWEB,“ 8th Biannual Conference on Solar Energy in High Latitudes(North Sun’99), Accessed: Jul. 03, 2020. [Online]. Available: (<https://www.osti.gov/etdweb/biblio/5956836>).
- Carpenter S. and Meloche N., The RETScreen Model for Simulating the Performance of Solar Air Heating Systems (Conference) | ETDEWEB,“ in Canadian Conference on Building Energy Simulation, Montreal, PQ (Canada), Montreal, 2002. Accessed: Jul. 05, 2020. [Online]. Available: (<https://www.osti.gov/etdweb/biblio/20309789>).
- Collins, M.R., Abulkhair, H., 2014. An evaluation of heat transfer and effectiveness for unglazed transpired solar air heaters. *Sol. Energy* vol. 99, 231–245. <https://doi.org/10.1016/J.SOLENER.2013.11.012>.
- Croitoru, C.V., Nastase, I., Bode, F.I., Meslem, A., 2016. Thermodynamic investigation on an innovative unglazed transpired solar collector. *Sol. Energy* vol. 131, 21–29. <https://doi.org/10.1016/J.SOLENER.2016.02.029>.
- Dabbas, A., Awana, A., Al Dabbas, A., Al-Hyari, L., Al Awana, A.D.M., Fawaiier, M., 2021. Energetic investigation and economic feasibility for a University campus in Romania Towards Becoming an Energy Supplier. *pp. bme. huA Al Dabbas, L Al-Hyari, ADM Al Awana, M FawaiierPeriodica Polytech. Mech. Eng.*, 2021•pp. bme. hu vol. 65 (2), 180–188. <https://doi.org/10.3311/PPme.17361>.
- C. Dymond, C. K.-S. Energy, and undefined 1997, Development of A Flow Distribution and Design Model for Transpired Solar Collectors, Elsevier, Accessed: Sep. 25, 2023. [Online]. Available: (<https://www.sciencedirect.com/science/article/pii/S0038092X96001570>).
- Elsarrag, E., Al-Horr, Y., Imbabi, M.S.-E.E., Salah-Eldin Imbabi, M., Imbabi, M.S.-E.E., 2012. Improving building fabric energy efficiency in hot-humid climates using dynamic insulation. *Build. Simul.* 2012 5:2 vol. 5 (2), 127–134. <https://doi.org/10.1007/S12273-012-0067-6>.
- Eryener, D., Hollick, J., Kuscü, H., 2017. Thermal performance of a transpired solar collector updraft tower. *Energy Convers. Manag.* vol. 142, 286–295. <https://doi.org/10.1016/J.ENCONMAN.2017.03.052>.
- Eryener, D., Kuscü, H., 2018. Hybrid transpired solar collector updraft tower. *Sol. Energy* vol. 159, 561–571. <https://doi.org/10.1016/j.solener.2017.11.035>.
- Fawaiier, Mohammad, Bokor, Balázs, 2020. Wall Heat Recapture with Transpired Solar Collectors in European Cities, Request PDF. 28. *medzinárod. vedecko-Odb. Konf.* 241–248. (https://www.researchgate.net/publication/342426067_WALL_HEAT_RECAPTURE_WITH_TRANSPICRED_SOLAR_COLLECTORS_IN_EUROPEAN_CITIES). Accessed: Jun. 29, 2020. [Online]. Available:
- M. Fawaiier and B. Bokor, Dynamic Insulation Systems of Building Envelopes | Request PDF,“ in *Vykurovanie 2021 Alternatívne zdroje energie pre budovy s takmer nulovou potrebou energie*, Stará Ľubovňa - Slovakia, 2021. Accessed: May 10, 2022. [Online]. Available: (https://www.researchgate.net/publication/358190193_DYNAMIC_INSULATION_SYSTEMS_OF_BUILDING_ENVELOPES).
- Fawaiier, M., Bokor, B., 2022. Dynamic insulation systems of building envelopes: a review. *Energy Build.* vol. 270, 112268 <https://doi.org/10.1016/J.ENBUILD.2022.112268>.
- Fawaiier, M., Bokor, B., Horváth, M., 2021. Wall heat loss recapture evaluation of transpired solar collectors for different climates: a european case study. *Case Stud. Therm. Eng.* vol. 24, 100836 <https://doi.org/10.1016/j.csite.2020.100836>.
- Federal Technology Alert, Transpired Collectors (Solar Preheaters for Outdoor Ventilation Air), 1998. Accessed: Jun. 24, 2020. [Online]. Available: (https://www.energy.gov/sites/prod/files/2013/10/F3/FTA_trans_coll.pdf).
- Fawaiier, Mohammad, Shaban, Nabeel Abu, Balázs, Bokor, 2024. Heat transfer investigation for wall heat loss recapture in transpired solar collectors. *Energy Conversion and Management: X* 22, 100540. <https://doi.org/10.1016/j.ecmx.2024.100540>.
- Filimonova, I.V., Nemov, V.Y., Komarova, A.V., Mishenin, M.V., Kozhev, V.D., 2021. Relationship of renewable energy consumption to economic, environmental and institutional factors in Europe. *Energy Rep.* vol. 7, 358–365. <https://doi.org/10.1016/J.EGYR.2021.07.115>.
- Fluent, A., 2020. *Ansys fluent. Acad. Res. Release* vol. 14.
- K. Gawlik, A Numerical and Experimental Investigation of Heat Transfer Issues in the Practical Utilization of Unglazed, Transpired Solar Air Heaters. 1995. Accessed: Sep. 25, 2023. [Online]. Available: (<https://search.proquest.com/openview/4da4936f7d65bf85102b5292586de212/1?pq-origsite=gscholar&cbl=18750&diss=y>).
- Gawlik, K., Christensen, C., Kutscher, C., 2005. A numerical and experimental investigation of low-conductivity unglazed, transpired solar air heaters. *J. Sol. Energy Eng., Trans. ASME* vol. 127 (1), 153–155. <https://doi.org/10.1115/1.1823494>.
- Gawlik, K.M., Kutscher, C.F., 2002. Wind heat loss from corrugated, transpired solar collectors. *J. Sol. Energy Eng.* vol. 124 (3), 256–261. <https://doi.org/10.1115/1.1487886>.
- Gholampour, M., Ameri, M., 2016. Energy and exergy analyses of Photovoltaic/Thermal flat transpired collectors: experimental and theoretical study. *Appl. Energy* vol. 164, 837–856. <https://doi.org/10.1016/J.APENERGY.2015.12.042>.
- A. Golneshan, Forced Convection Heat Transfer from Low Porosity Slotted Transpired Plates, 1996, Accessed: Oct. 31, 2023. [Online]. Available: (<https://elibrary.ru/item.asp?id=5614225>).
- Gunnawick, L.H., Brundrett, E., Hollands, K.G.T., 1996. Flow distribution in unglazed transpired plate solar air heaters of large area. *Sol. Energy* vol. 58 (6), 221–237.
- Hollick, J.C., 1994. Unglazed solar wall air heaters. *Renew. Energy* vol. 5 (1–4), 415–421. [https://doi.org/10.1016/0960-1481\(94\)90408-1](https://doi.org/10.1016/0960-1481(94)90408-1).
- Imbabi, M.S.E., 2006. Modular breathing panels for energy efficient, healthy building construction. *Renew. Energy* vol. 31 (5), 729–738. <https://doi.org/10.1016/J.RENENE.2005.08.009>.
- Industrial Perforators Association, Handbook for Perforated Metals, Houston, Texas, 1993. Accessed: Oct. 05, 2023. [Online]. Available: (<https://www.marcospecialtysteel.com/content/uploads/2017/01/Perforated-Handbook-for-designers-min.pdf>).
- S.J.M. Koenders, Active Insulation Computational assessment of the performance of a forced convective dynamic insulation system / Department of the Built Environment Unit Building Physics and Services, Eindhoven University of Technology, Eindhoven, Netherlands, 2018. Accessed: Nov. 13, 2021. [Online]. Available: (https://pure.tue.nl/ws/portals/portal/97347481/Koenders_0922583.pdf).
- W.J. Krotiuk, Development of a Pressure Drop Calculation Method for Debris-covered Sump Screens in Support of Generic Safety Issue 191. Division of Risk Assessment and Special Projects, Office of Nuclear ..., 2007.
- C.F. Kutscher, An Investigation of Heat Transfer for Air Flow through Low-porosity Perforated Plates, University of Colorado at Boulder, 1992.
- Kutscher, C.F., 1994. Heat exchange effectiveness and pressure drop for air flow through perforated plates with and without crosswind. *J. Heat. Transf.* vol. 116 (2), 391–399. <https://doi.org/10.1115/1.2911411>.
- C. Kutscher, Transpired Solar Collector Systems: A Major Advance in Solar Heating, 1997, Accessed: Sep. 25, 2023. [Online]. Available: (<https://www.osti.gov/biblio/478238>).
- Kutscher, C.F., Christensen, C.B., Barker, G.M., 1993b. Unglazed transpired solar collectors: heat loss theory (Aug.). *J. Sol. Energy Eng. Trans. ASME* vol. 115 (3), 182–188. <https://doi.org/10.1115/1.2930047>.
- Kutscher, C., Christensen, C., Gawlik, K., 2003. Letter to the Editor. *Sol. Energy* vol. 74 (4), 353–354. [https://doi.org/10.1016/S0038-092X\(03\)00148-8](https://doi.org/10.1016/S0038-092X(03)00148-8).
- Kutscher, C.F., Christensen, C.B., Barker, G.M., Kutscher, C.F., Christensen, C.B., Barker, G.M., 1993a. Unglazed transpired solar collectors: heat loss theory (Aug.). *J. Sol. Energy Eng. Trans. ASME* vol. 115 (3), 182–188. <https://doi.org/10.1115/1.2930047>.
- Leon, M.A., Kumar, S., 2007. Mathematical modeling and thermal performance analysis of unglazed transpired solar collectors. *Sol. Energy* vol. 81 (1), 62–75. <https://doi.org/10.1016/j.solener.2006.06.017>.
- S. Li and P. Karava, Numerical Study of Convective Heat Transfer for Flat Unglazed Transpired Solar Collectors, 2012, Accessed: Oct. 30, 2023. [Online]. Available: (<https://docs.lib.purdue.edu/ihpbc/75/>).
- Li, S., Karava, P., Currie, S., Lin, W.E., Savory, E., 2014. Energy modeling of photovoltaic thermal systems with corrugated unglazed transpired solar collectors - Part 1: model development and validation. *Sol. Energy* vol. 102, 282–296. <https://doi.org/10.1016/j.solener.2013.12.040>.
- Natural Resources Canada, Clean Energy Project Analysis Third Edition RETScreen® Engineering & Cases Textbook. Accessed: Jul. 03, 2020. [Online]. Available: (<http://msessd.ioe.edu.np/wp-content/uploads/2017/04/Textbook-clean-energy-project-analysis.pdf>).
- NRCan, “RETScreen Clean Energy Project Analysis - Engineering & Cases Textbook,” 2005. Accessed: Jul. 02, 2020. [Online]. Available: (<http://msessd.ioe.edu.np/wp-content/uploads/2017/04/Textbook-clean-energy-project-analysis.pdf>).
- Pop, O., Berville, C., Bode, F., Croitoru, C., 2022. Numerical investigation of cascaded phase change materials use in transpired solar collectors. *Energy Rep.* vol. 8, 184–193. <https://doi.org/10.1016/J.EGYR.2022.06.114>.

- RETScreen | Natural Resources Canada.” Accessed: Dec. 29, 2019. [Online]. Available: (<https://www.nrcan.gc.ca/maps-tools-publications/tools/data-analysis-software-modelling/retscreen/7465>).
- Saidi, K., Hammami, S., 2015. The impact of CO₂ emissions and economic growth on energy consumption in 58 countries. *Energy Rep.* vol. 1, 62–70. <https://doi.org/10.1016/J.EGYR.2015.01.003>.
- T. Semenou, D. Rousse, ... B. L. L.-M., and undefined 2015, “Mathematical modeling of dual intake transparent transpired solar collector,” *hindawi.com* T. Semenou, DR Rousse, B Le Lostec, HF Nouanegue, PL Paradis *Mathematical problems in engineering*, 2015 • *hindawi.com*, Accessed: Oct. 05, 2023. [Online]. Available: (<https://www.hindawi.com/journals/mpe/2015/942854/abs/>).
- Tajdaran, S., Bonatesta, F., Ogden, R., Kendrick, C., 2016. CFD modeling of transpired solar collectors and characterisation of multi-scale airflow and heat transfer mechanisms. *Sol. Energy* vol. 131, 149–164. <https://doi.org/10.1016/J.SOLENER.2016.02.042>.
- Taylor, B.J., Webster, R., Imbabi, M.S., 1998. The building envelope as an air filter. *Build. Environ.* vol. 34 (3), 353–361. [https://doi.org/10.1016/S0360-1323\(98\)00017-1](https://doi.org/10.1016/S0360-1323(98)00017-1).
- Van Decker, G.W.E., Hollands, K.G.T., Brunger, A.P., 2001. Heat-exchange relations for unglazed transpired solar collectors with circular holes on a square or triangular pitch (Jan.). *Sol. Energy* vol. 71 (1), 33–45. [https://doi.org/10.1016/S0038-092X\(01\)00014-7](https://doi.org/10.1016/S0038-092X(01)00014-7).
- Wu, Z., Jin, Y., Huang, L., Zhang, N., Luo, X., Wang, X., 2023. Appraisal of indoor environment quality (IEQ) in nearly Zero energy Building (nZEB): a literature review. *Sci. Technol. Built Environ.* vol. 29 (3), 251–267. <https://doi.org/10.1080/23744731.2023.2176680>.